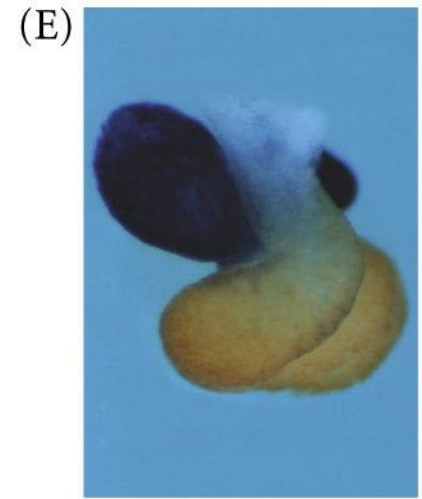
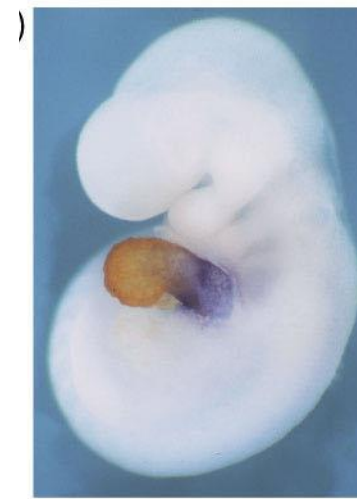
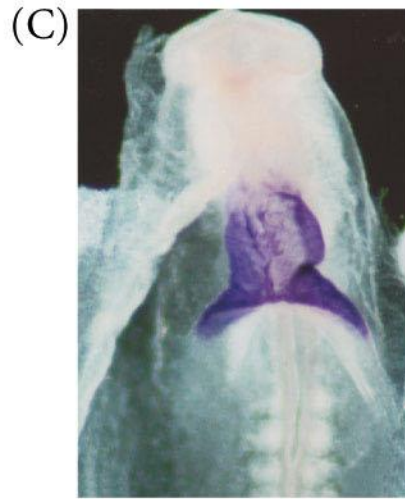
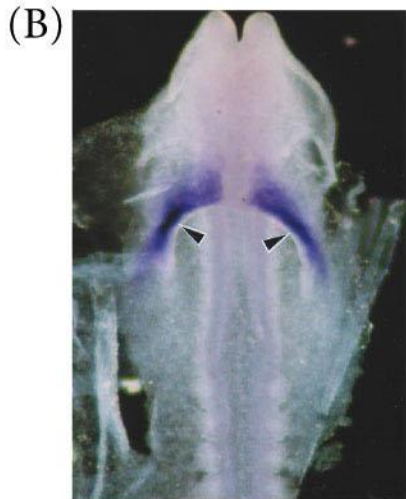


Development of the heart

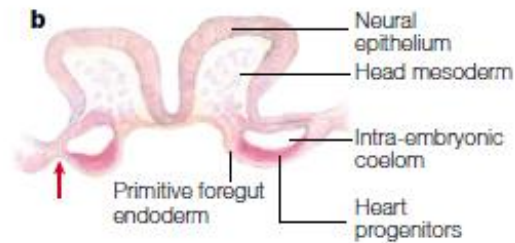
Dr. Nagy Nándor
Semmelweis University



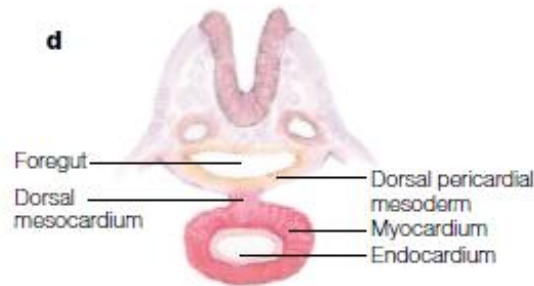
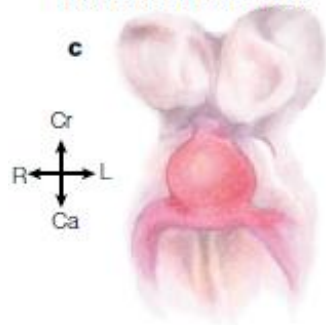
Keywords:

- heart field formation, cardiac mesoderm
- endocardium, myocardium, pericardium development
- cardiac loop, septum formation
- malformations

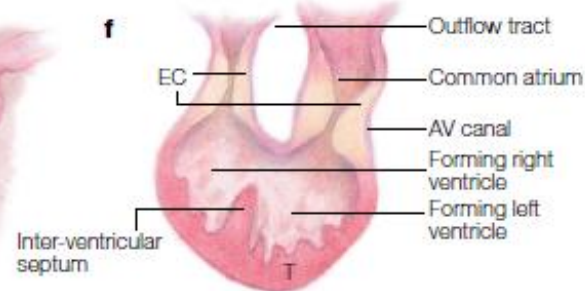
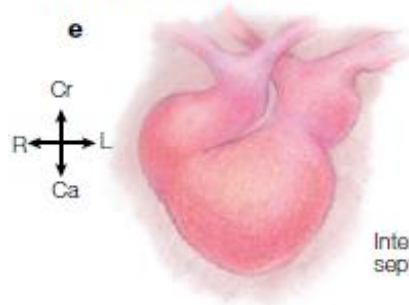
Cardiac crescent E7.75



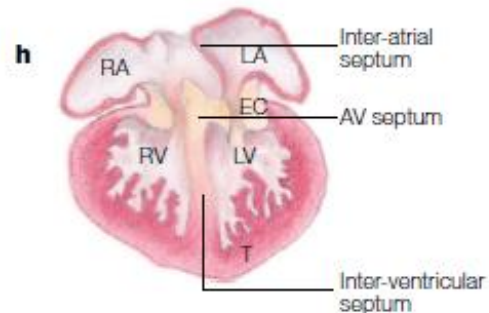
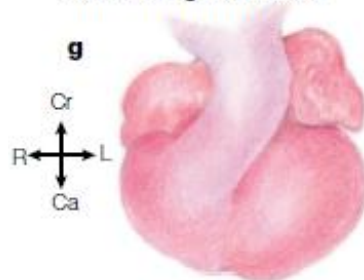
Linear heart tube E8.25



Looping heart E10.5



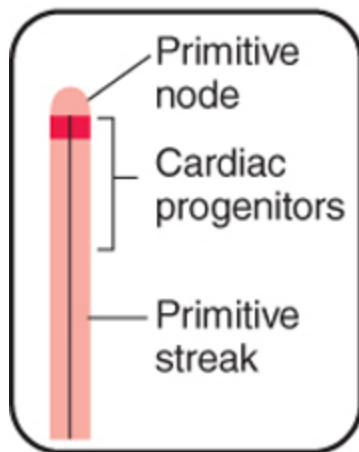
Remodelling heart E12.5



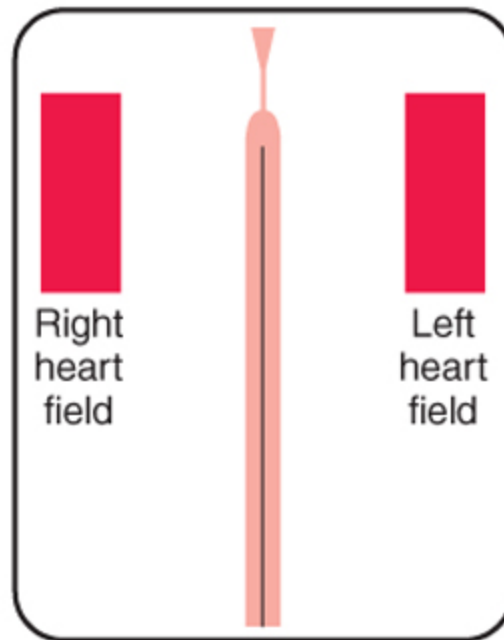
**Important
stages of the
heart
development**

Migration of the cardiac precursor cells:

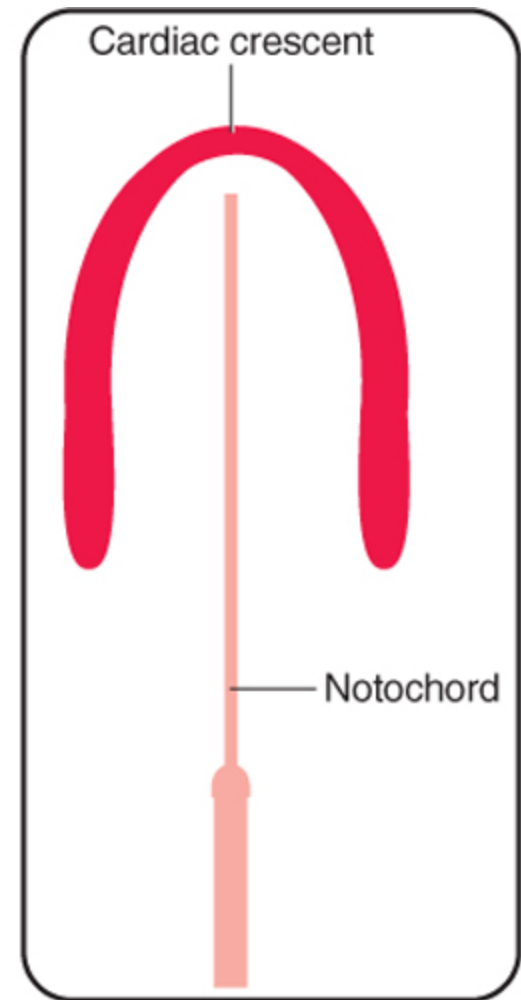
Presumptive heart cells **migrate anteriorly between the ectoderm and endoderm** toward the lateral walls of the anterior gut tube: forming the **CARDIAC CRESCENT**.



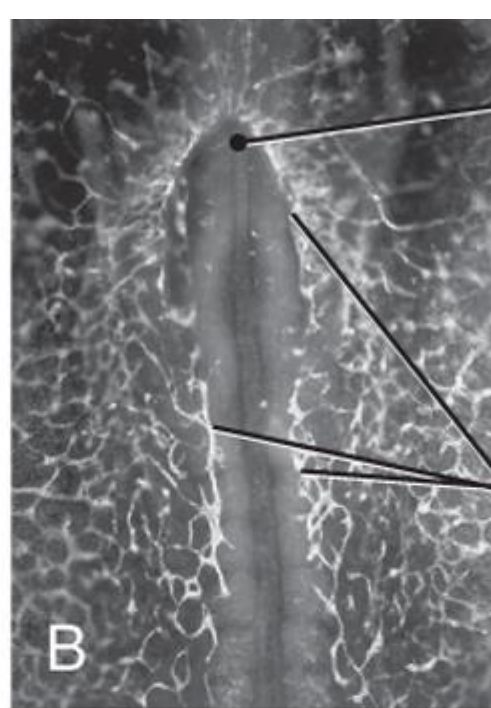
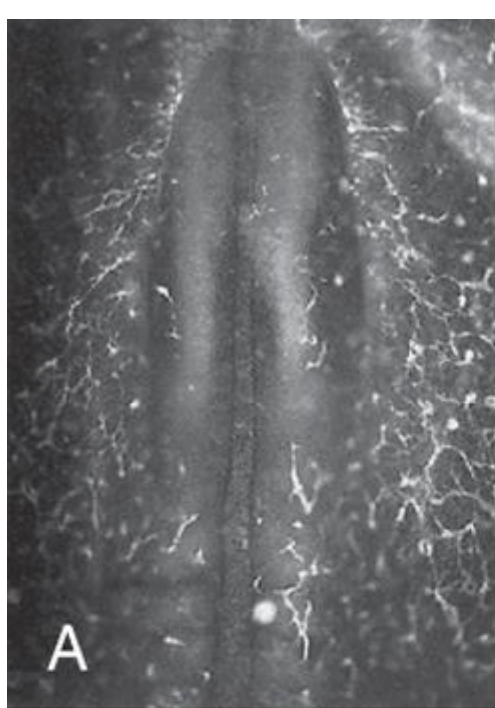
A



B



C

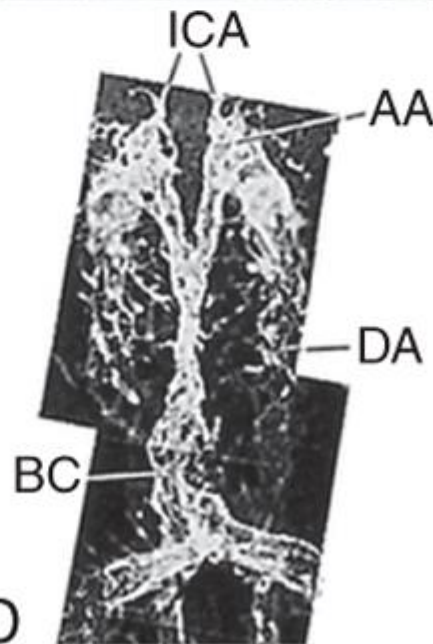
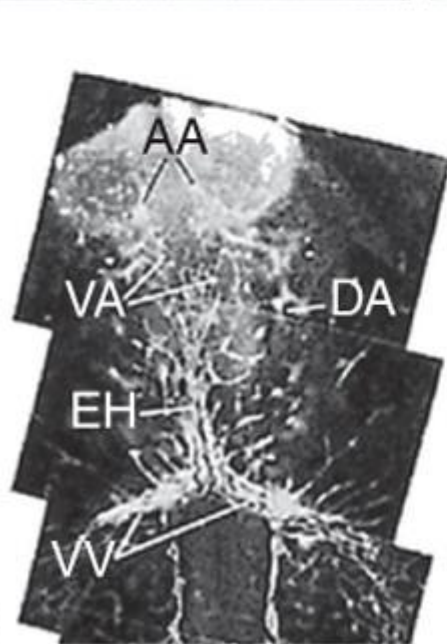


Anterior
intestinal
portal

Forming endo-
cardial
heart tubes

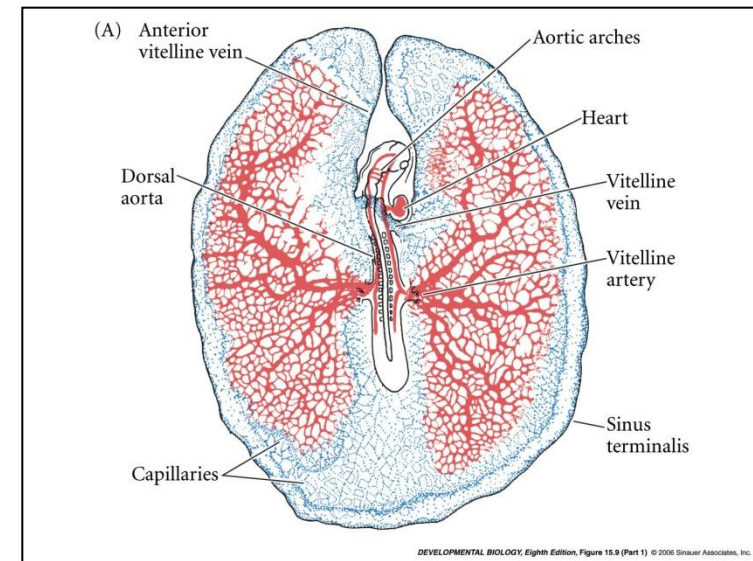
A

B

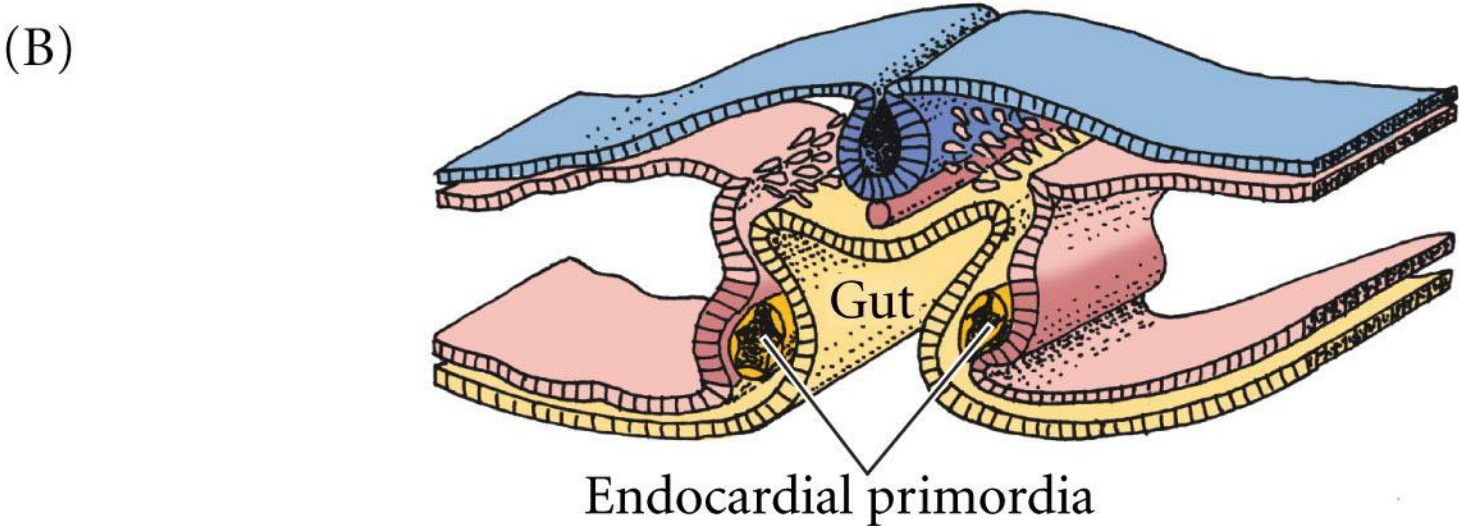
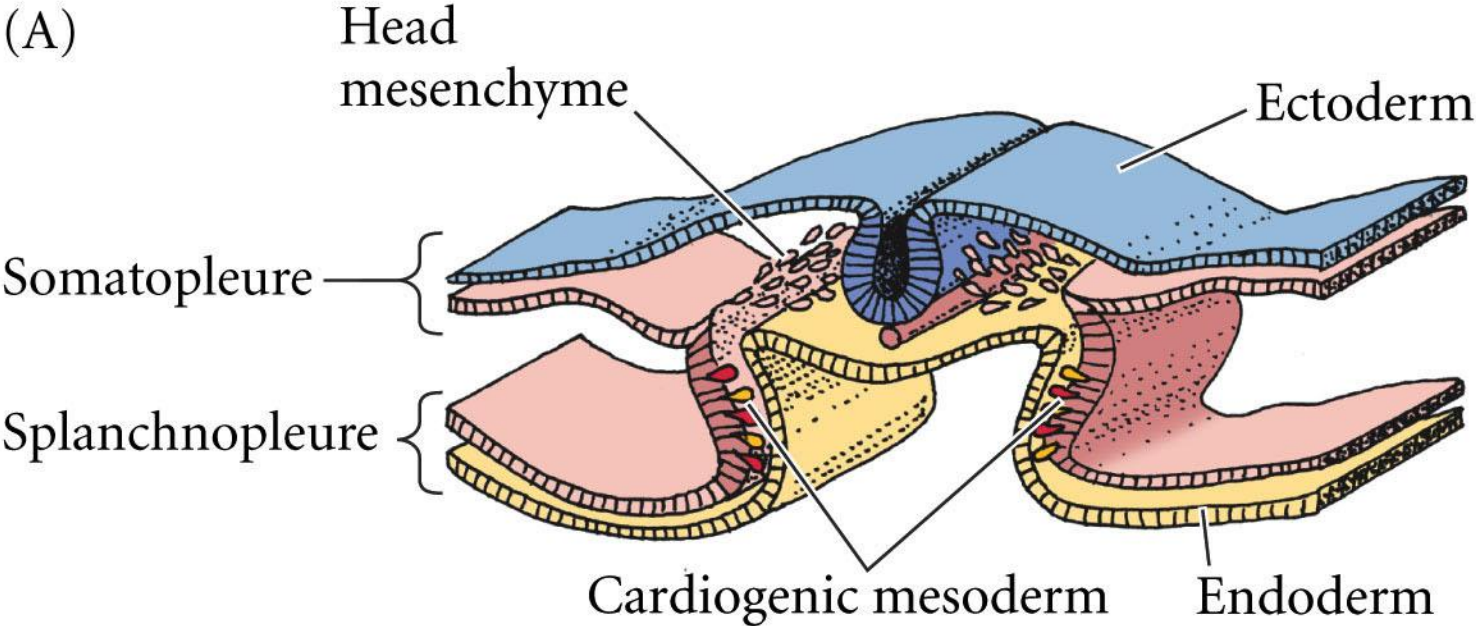


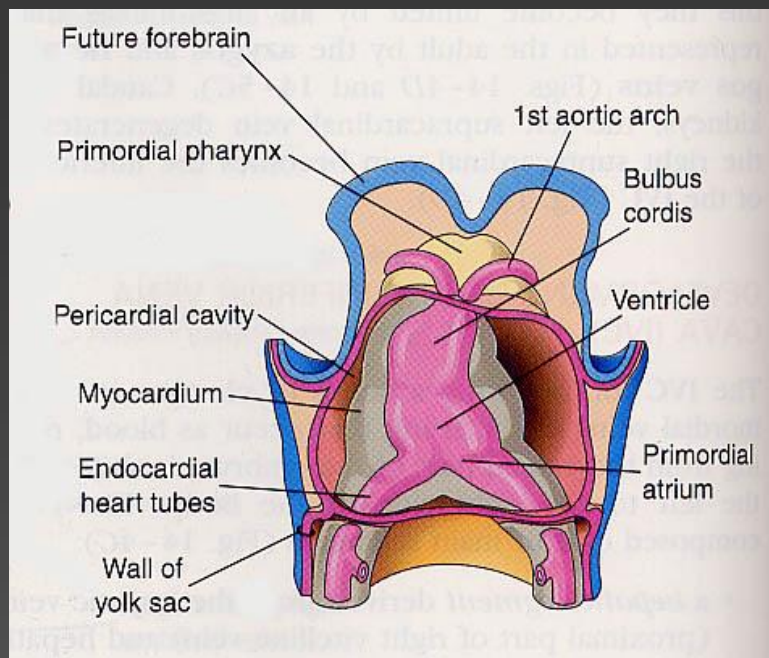
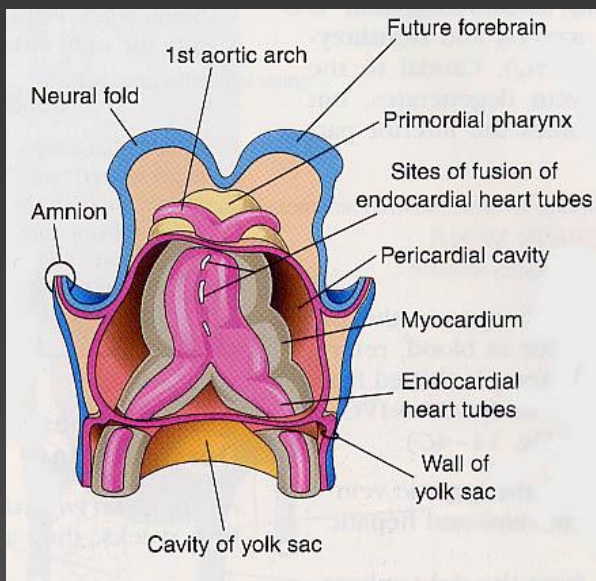
C

D

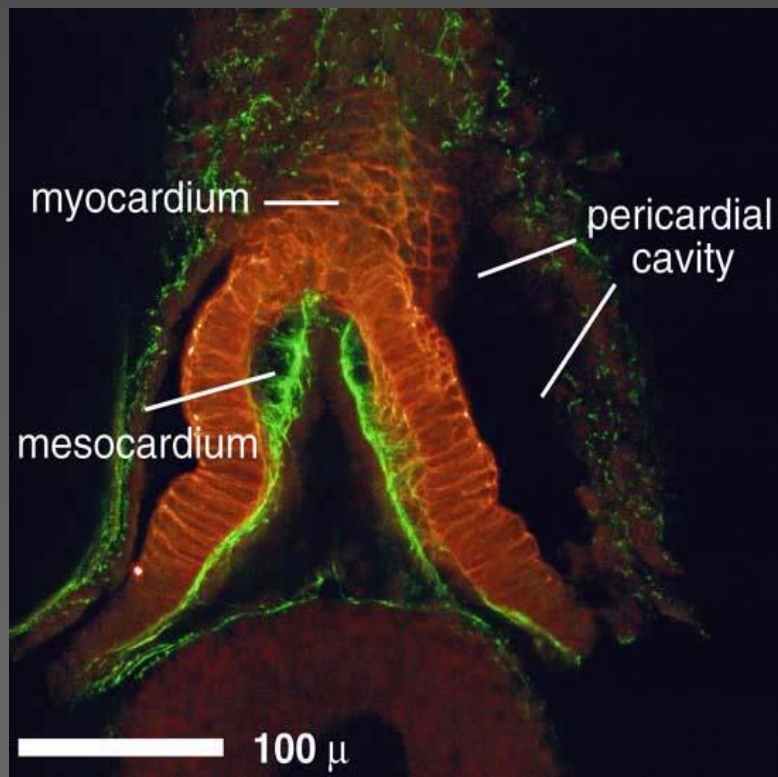


Formation of the heart from the splanchnic lateral plate mesoderm

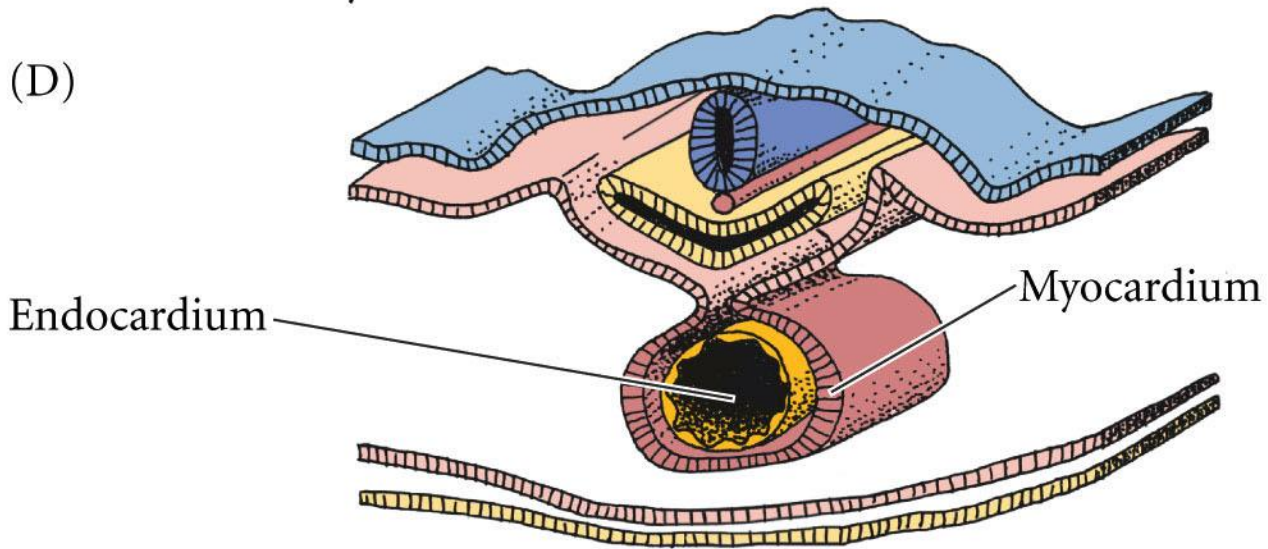
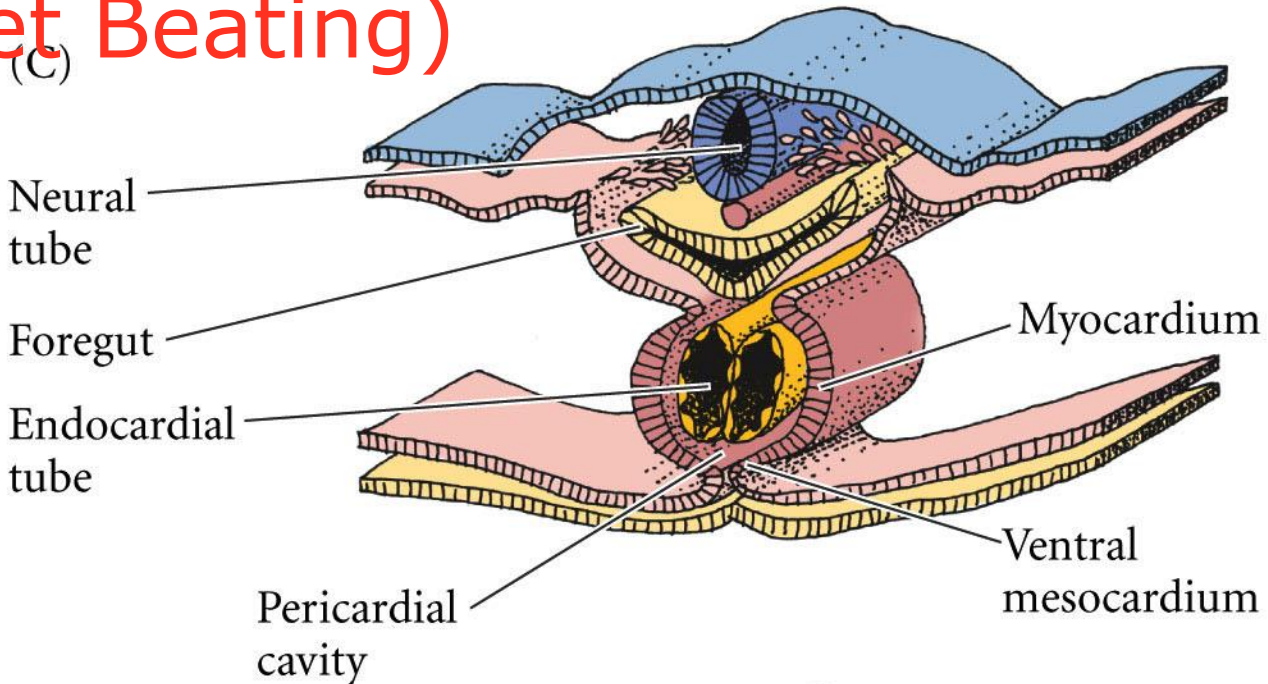


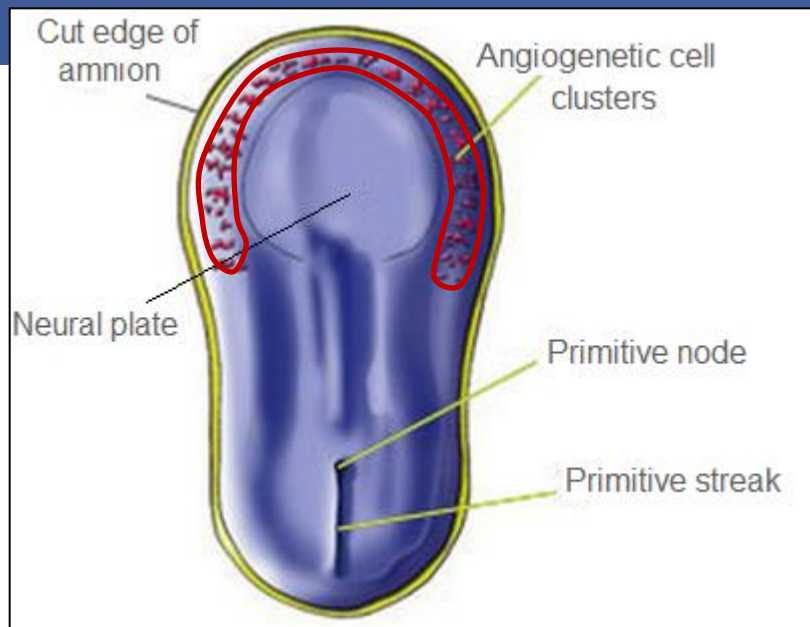


Ventral view of fusing endocardial tubes within pericardial cavity



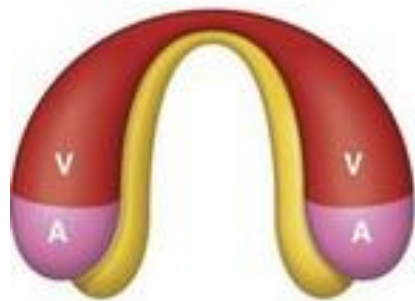
(Not Yet Beating)



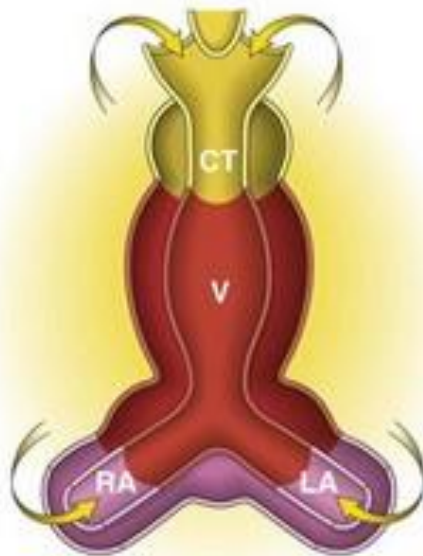


Within the cardiogenic area the **first** and **second heart** field appears (both of them have mesodermal origin). **First heart field** gives atria and the left ventricle, while from the **second one** the right ventricle and the conotruncal region (CT: bulbus cordis and truncus arteriosus) develops. **Immigrating neural crest cells also contribute to the formation of the aortic sac (AS).**

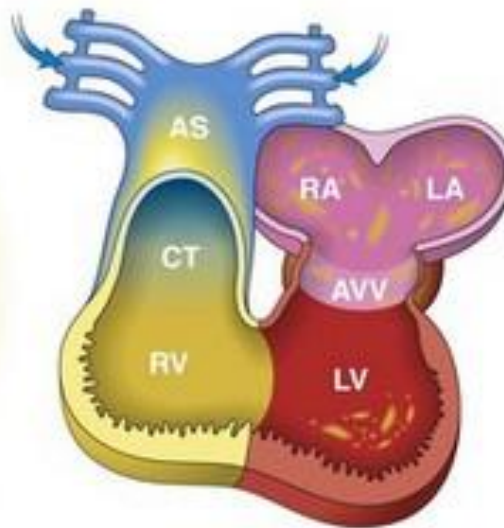
- First heart field
- Second heart field
- Cardiac neural crest



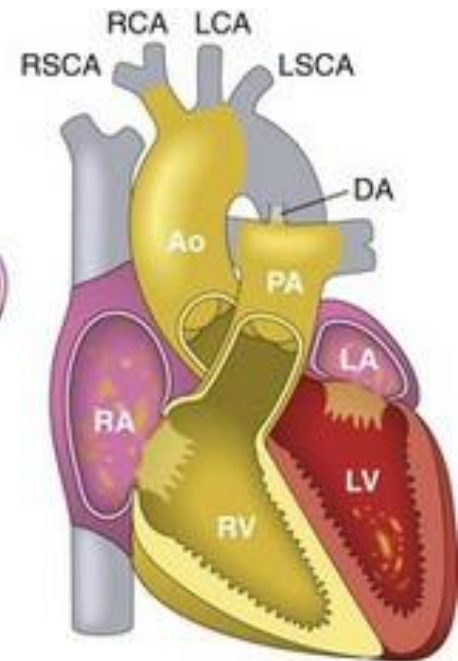
A First heart field template



B Second heart field migration



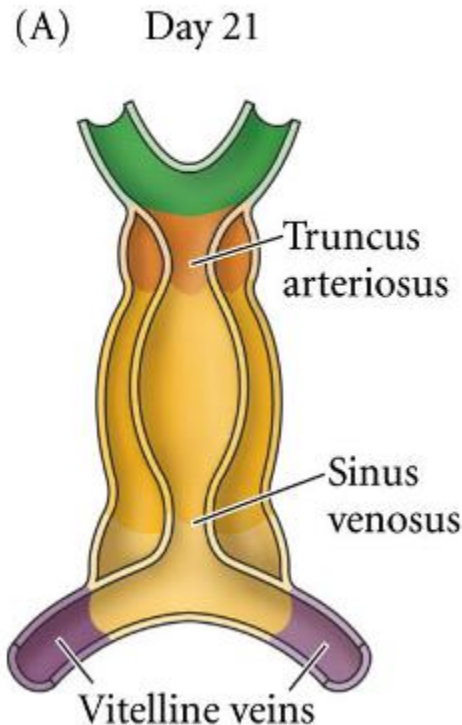
C Neural crest migration



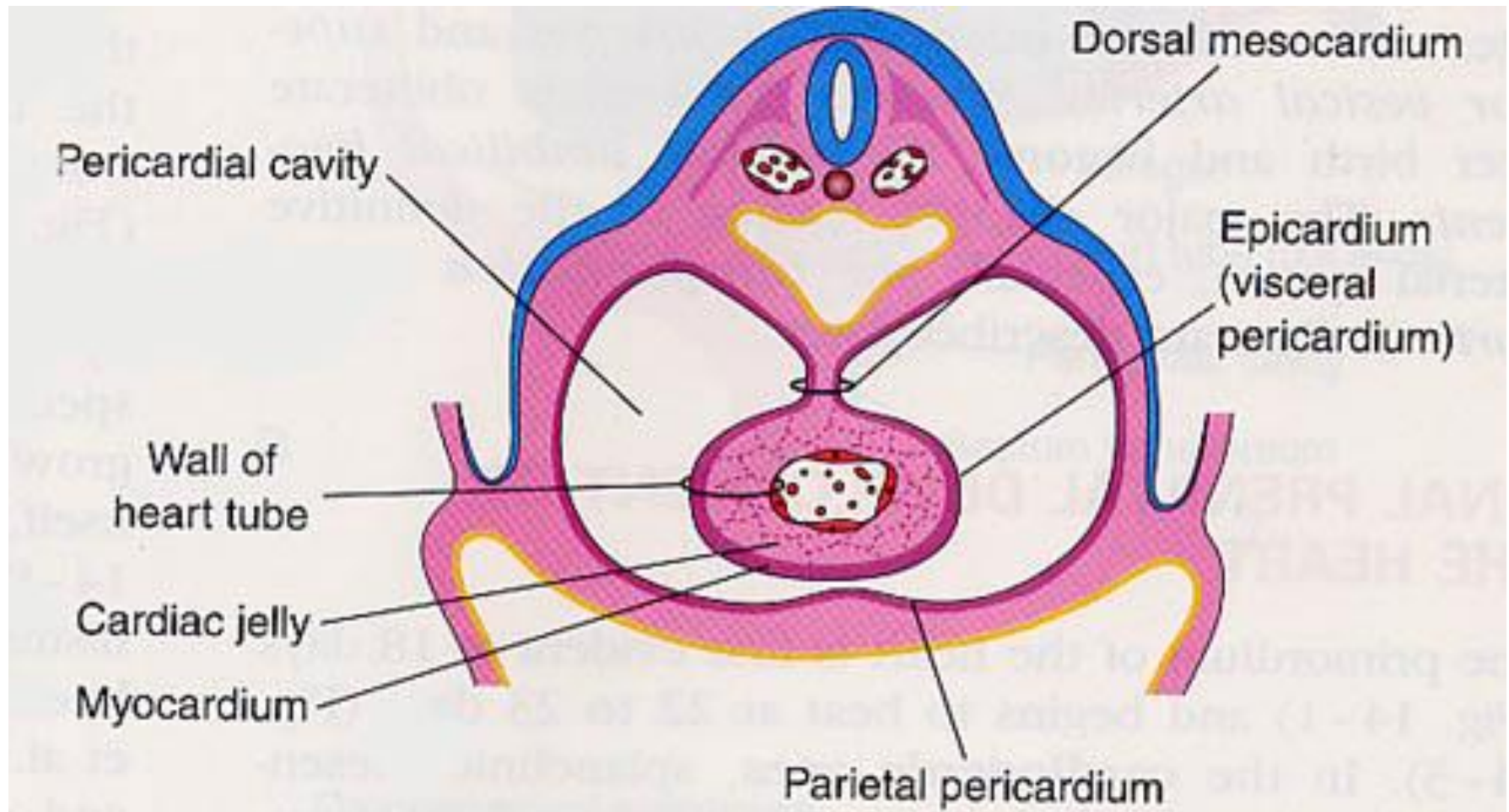
D Completed formation of four-chambered heart

Fusion of the heart rudiments and initial heartbeats

- Two myocardia unite to **form a single tube**
- **Vitelline veins**: the **unfused posterior portions of the endocardium**; carry nutrients from the yolk sac into the **sinus venosus**, the posterior region of the two fused veins
- **Truncus arteriosus**: contracts to speed the blood into the aorta
- **Pulsation**: initiate even during the paired primordia are still fusing by the heart muscle cells; $\text{Na}^+/\text{Ca}^{2+}$ exchange pump in the muscle cells executes
- **Rhythmicity**: coordinated **by the sinus venosus**, which generates electric impulses



Early Heart Development

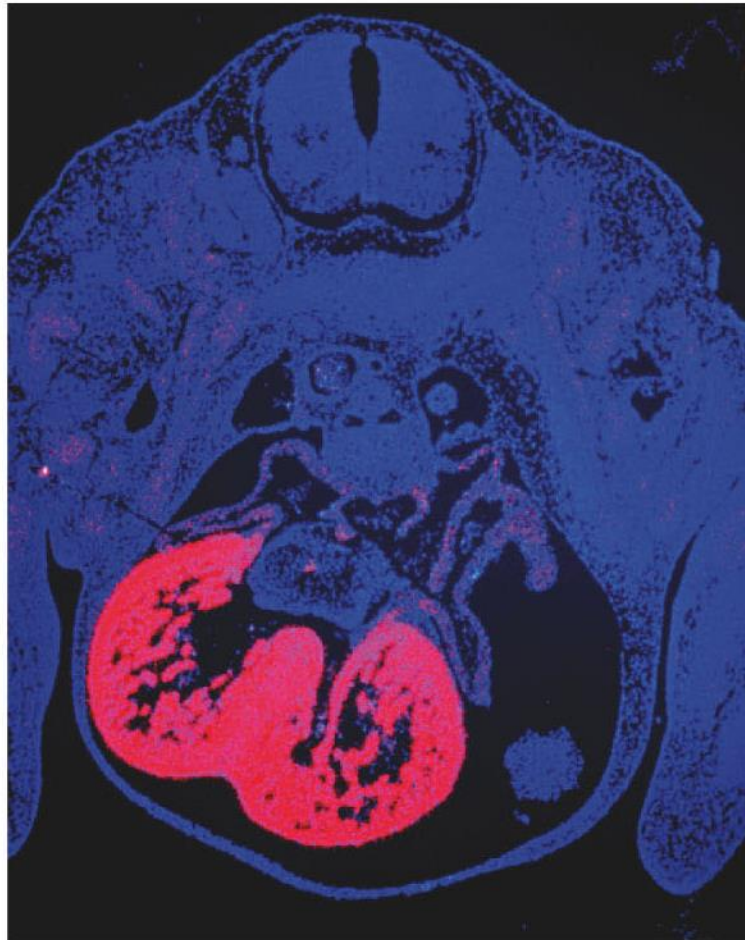


(Now it starts to Beat)

Thickening of the splanchnic mesoderm gives the myoepicardial mantles from which the MYOCARDIUM develops.

Migration of heart primordial: *cardia bifida*: two separate hearts

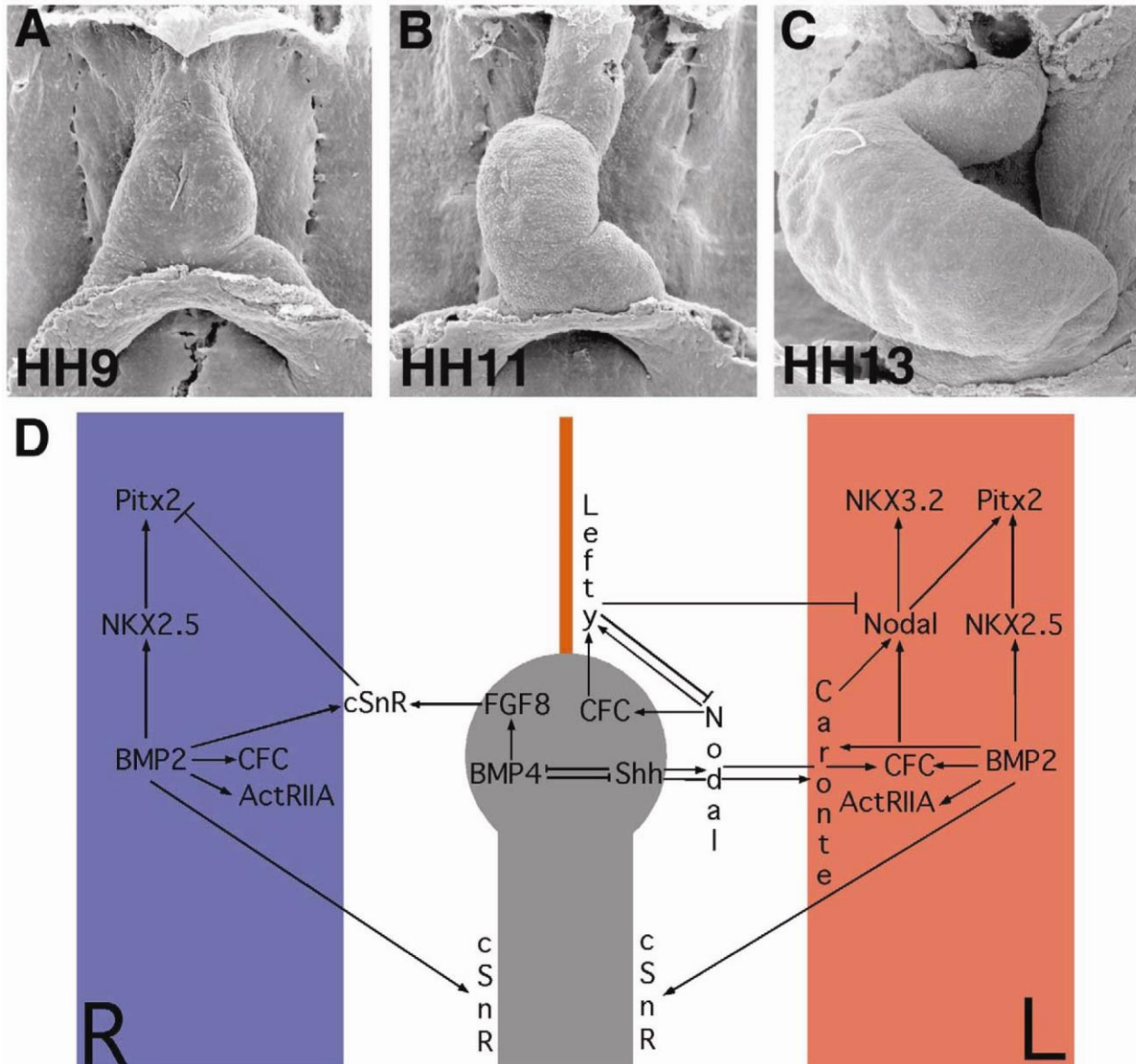
(D)



(E)



CARDIAC LOOP: left-right asymmetry



In early embryos (3-day chick or 5-week human), the heart is a two-chambered tube, through which blood comes in the lower chamber and being pumped out through the aorta

Looping changes the polarity of the heart from anterior-posterior axis to left-right axis: the right ventricle lies anterior to the left ventricle

Looping is dependent on the left-right patterning mediated by Nodal-Lefty2 signaling

Transcription factors: Hand1 becomes restricted to the left ventricle, while Hand2 to right

Other genes involve in differential growth of the heart

Tbx5 in left ventricle and artium

Wnt11 in the right side

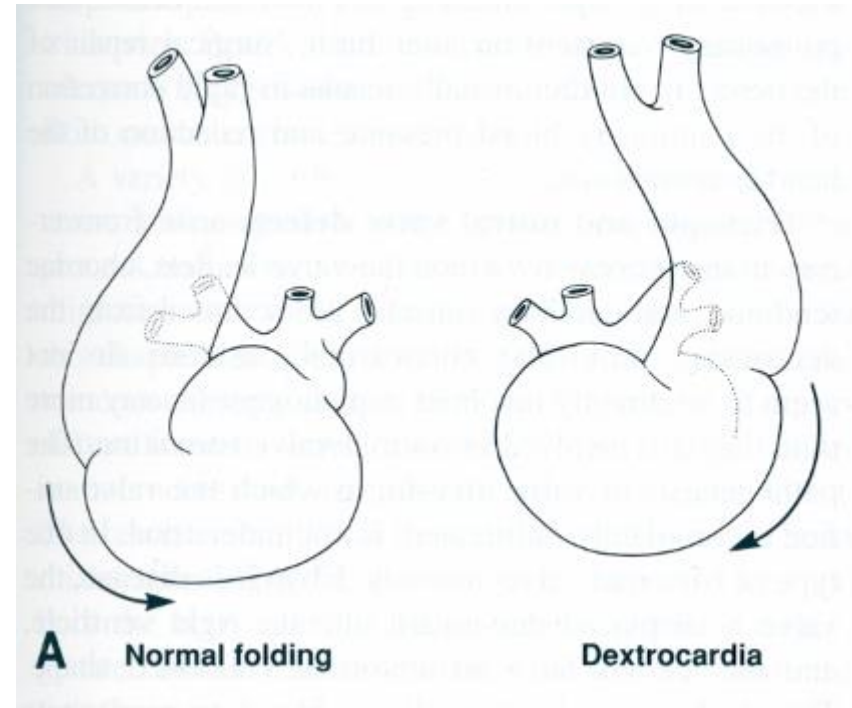
Four heart valves:

developed from the endocardial cushions dividing the atrioventricular channels; the endocardial cushions are being substituted by the specialized valves
atrium is partitioned by two septa

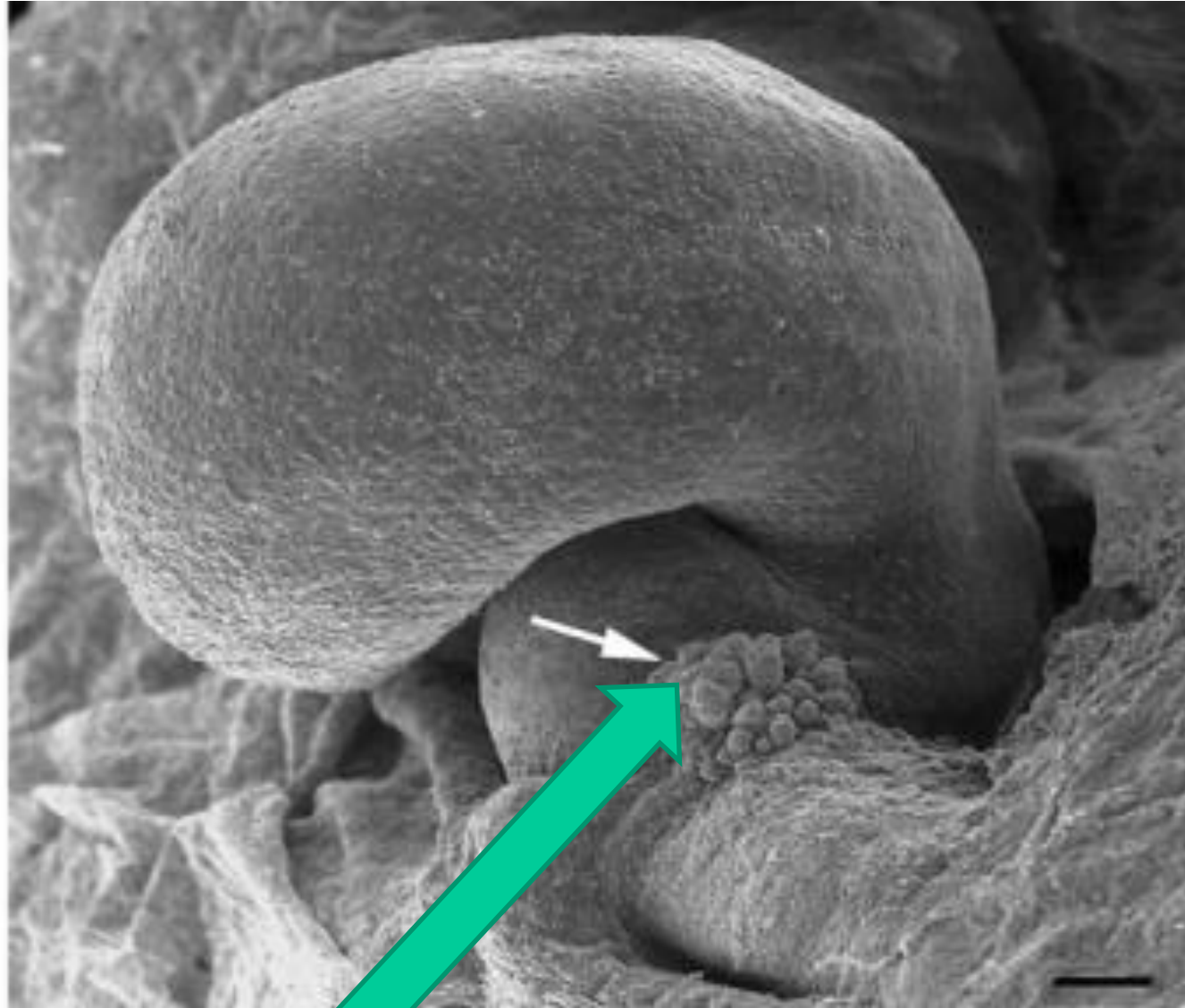
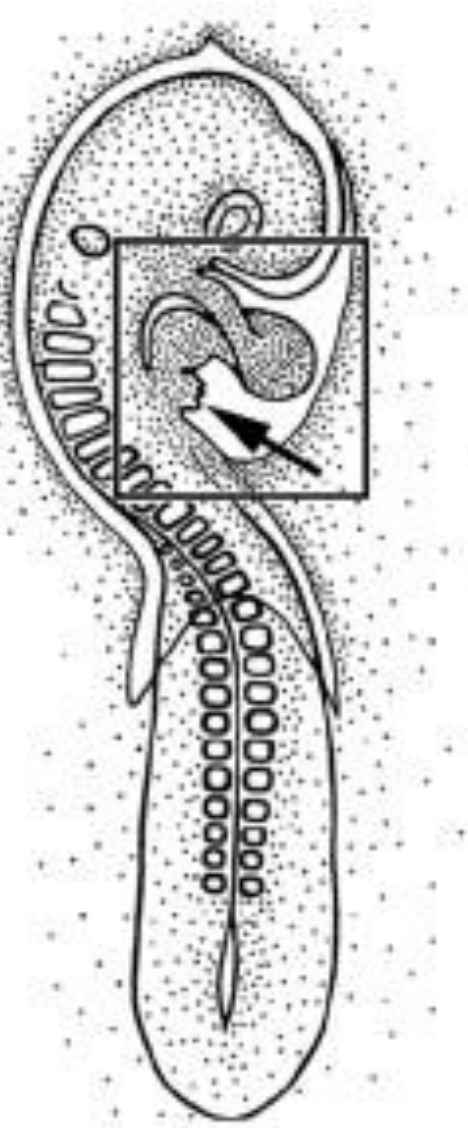


Dextrocardia; 1:7000

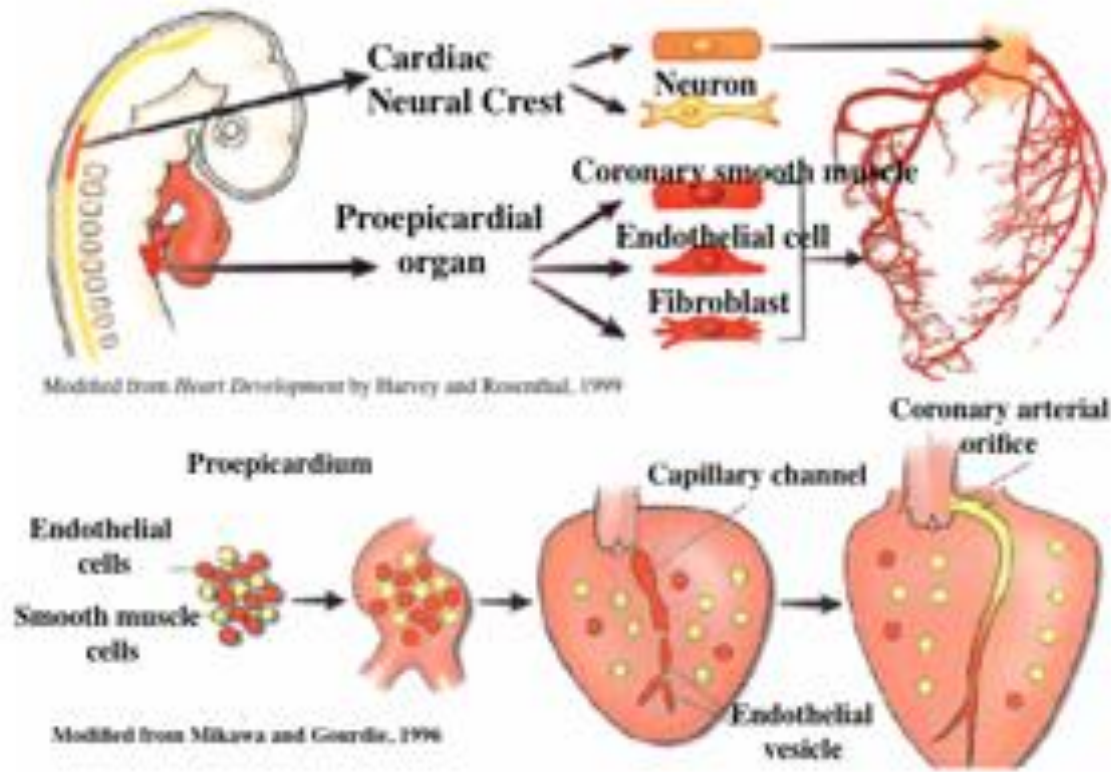
Dextrocardia is a congenital cardiac malrotation in which the heart is situated on the right side of the body (dextroversion) with the apex pointing to the right.



The **proepicardial organ** is a transient embryonic structure located just below the developing A/V canal of the early heart.



PROEPICARDIAL ORGAN



Schematic representations of CV development.

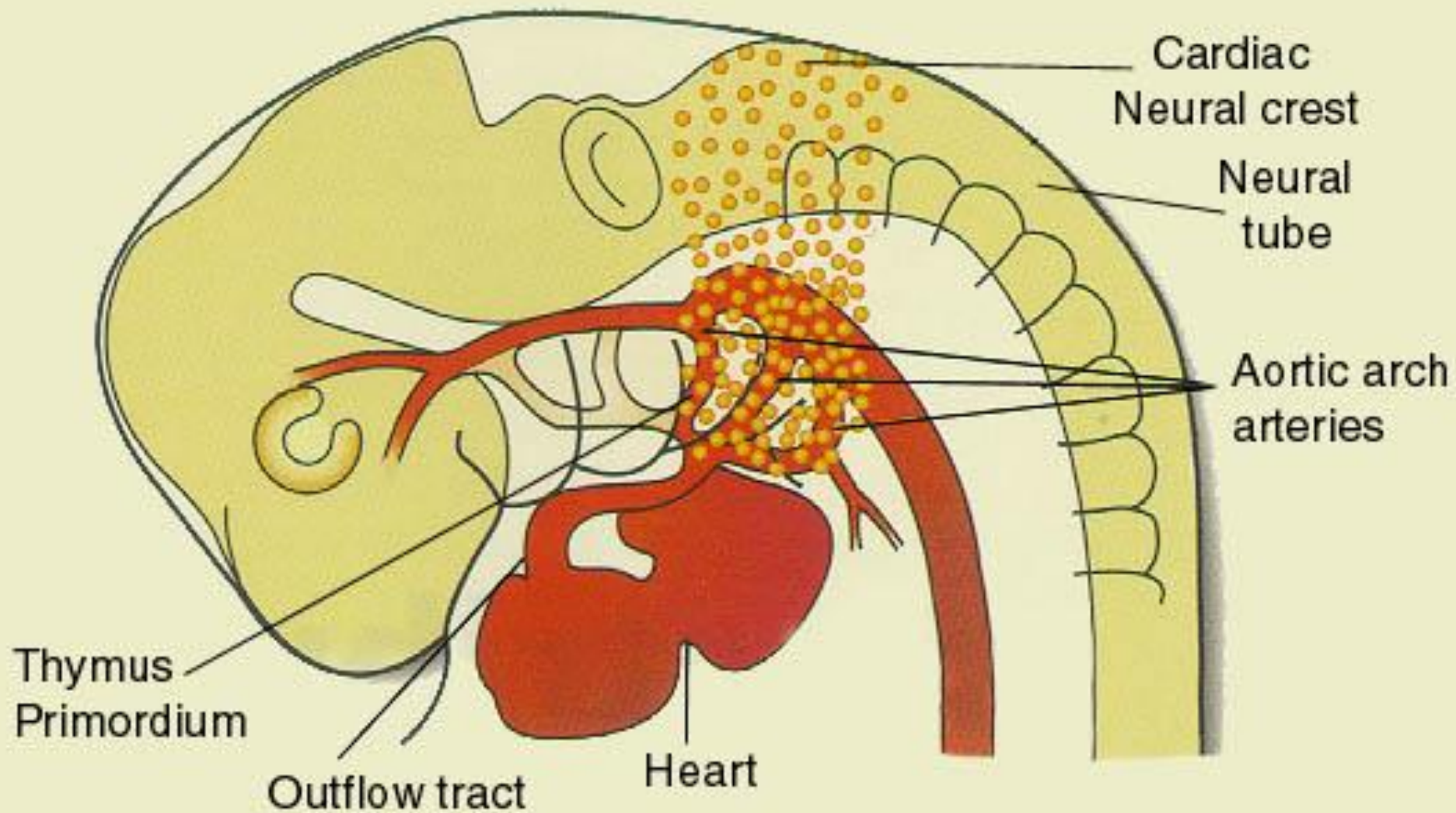
The PE is of central importance in CV development. Genetic labeling experiments in avian systems have established that this generative tissue contributes to all the vascular cells of the heart.

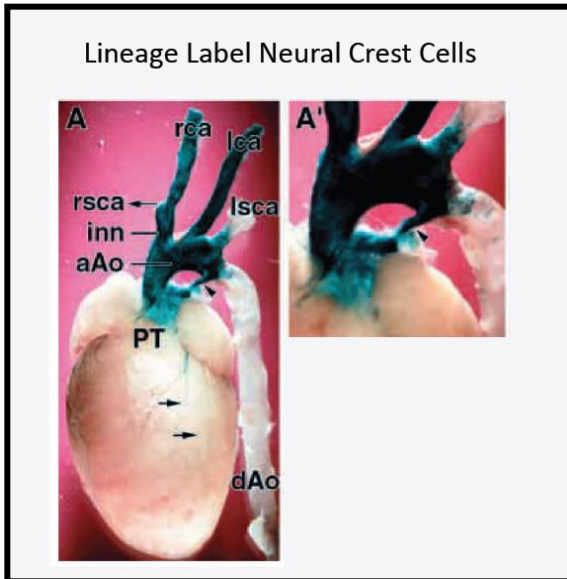
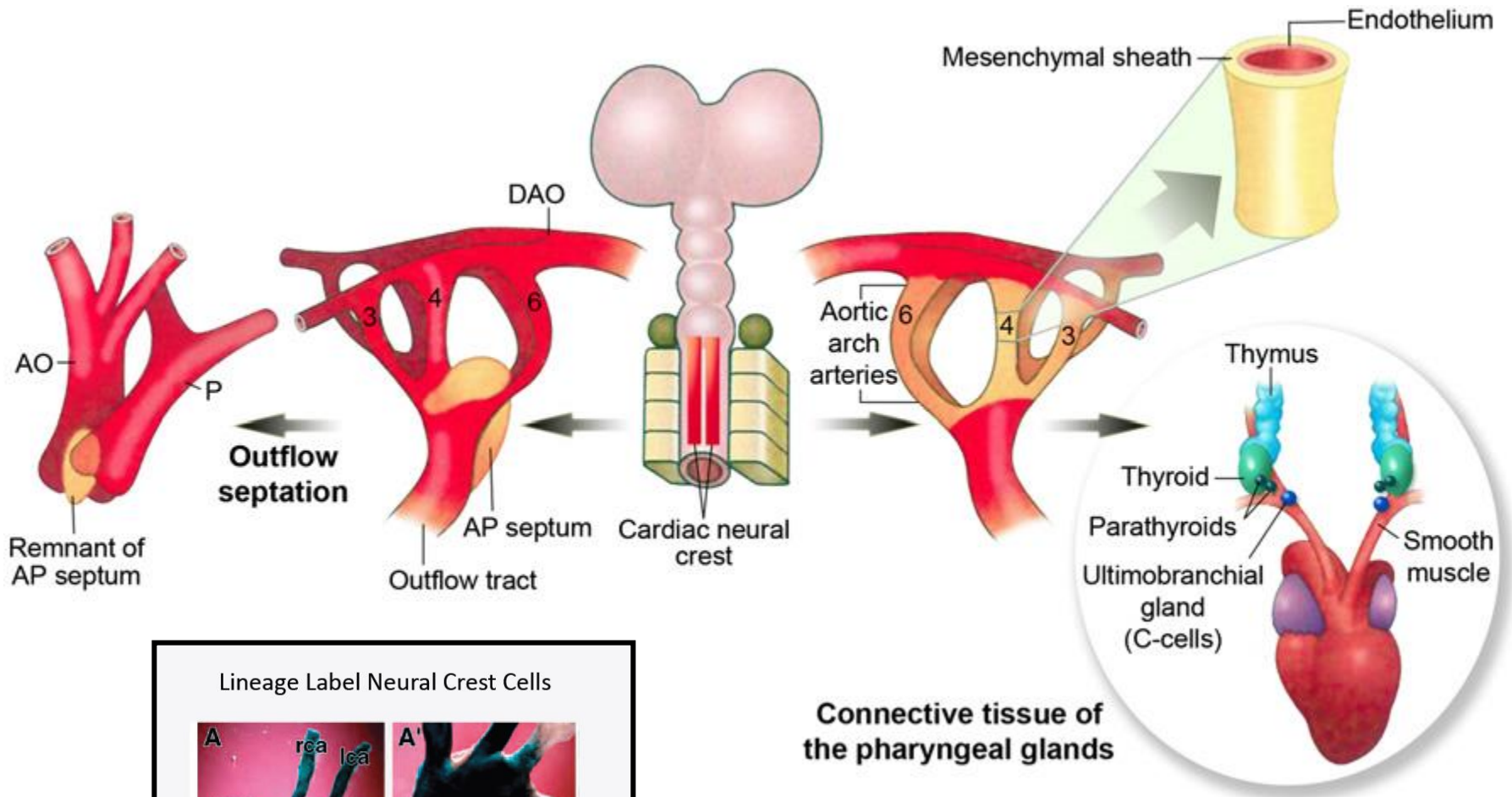
Cardiac neural crest

The **cardiac neural crest** overlaps the vagal neural crest and migrates to populate the pharyngeal arches 3, 4 and 6 (producing structures in the head) and to the heart, forming connective tissue that separates the great vessels of the heart.

Other Migration Locations:

Into the pharyngeal arches and Truncus arteriosus (embryology), forming the aorticopulmonary septum and the smooth muscle of great arteries.

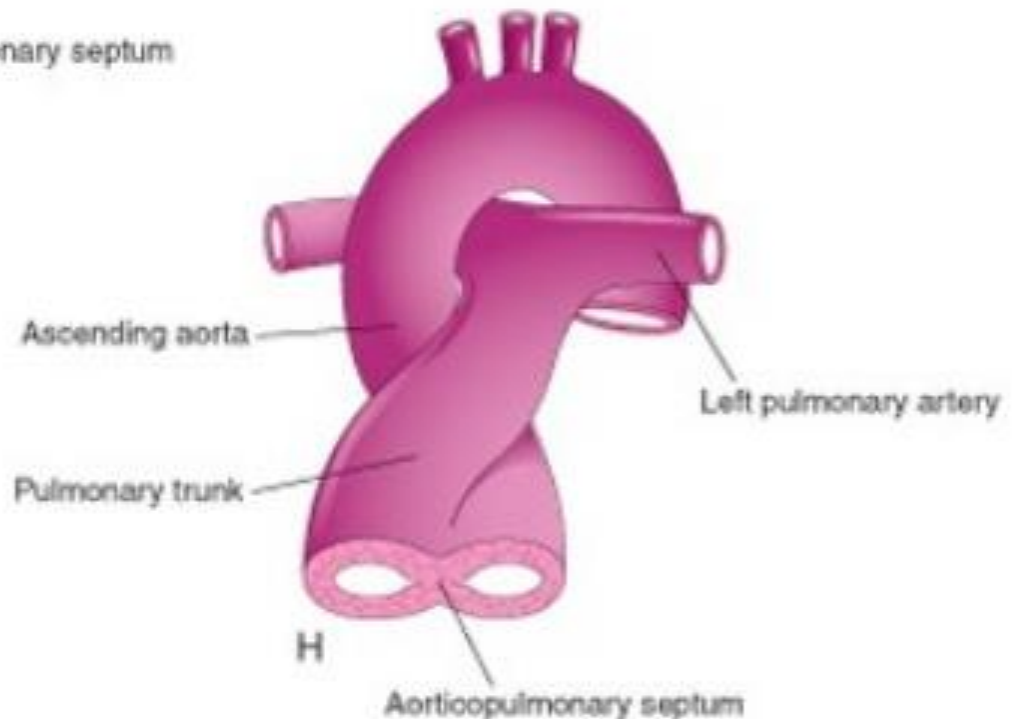
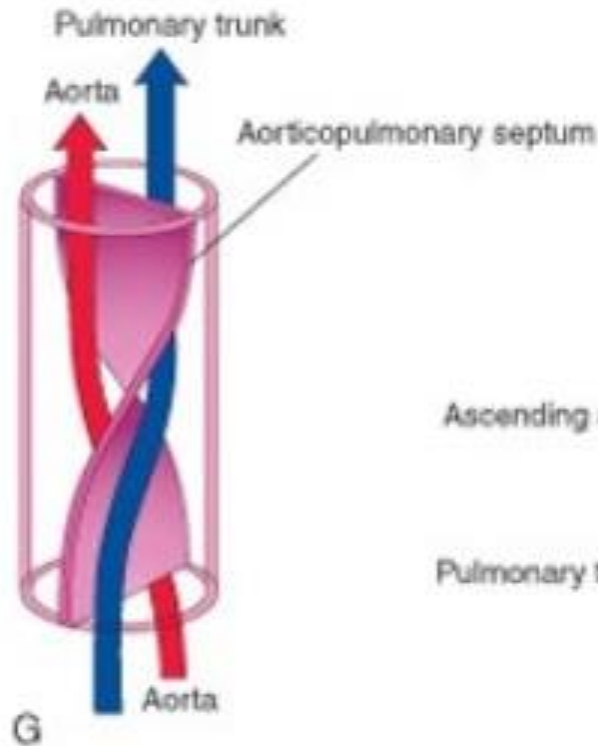
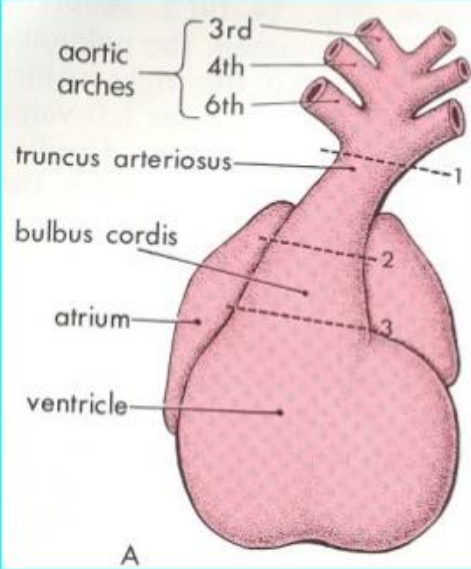




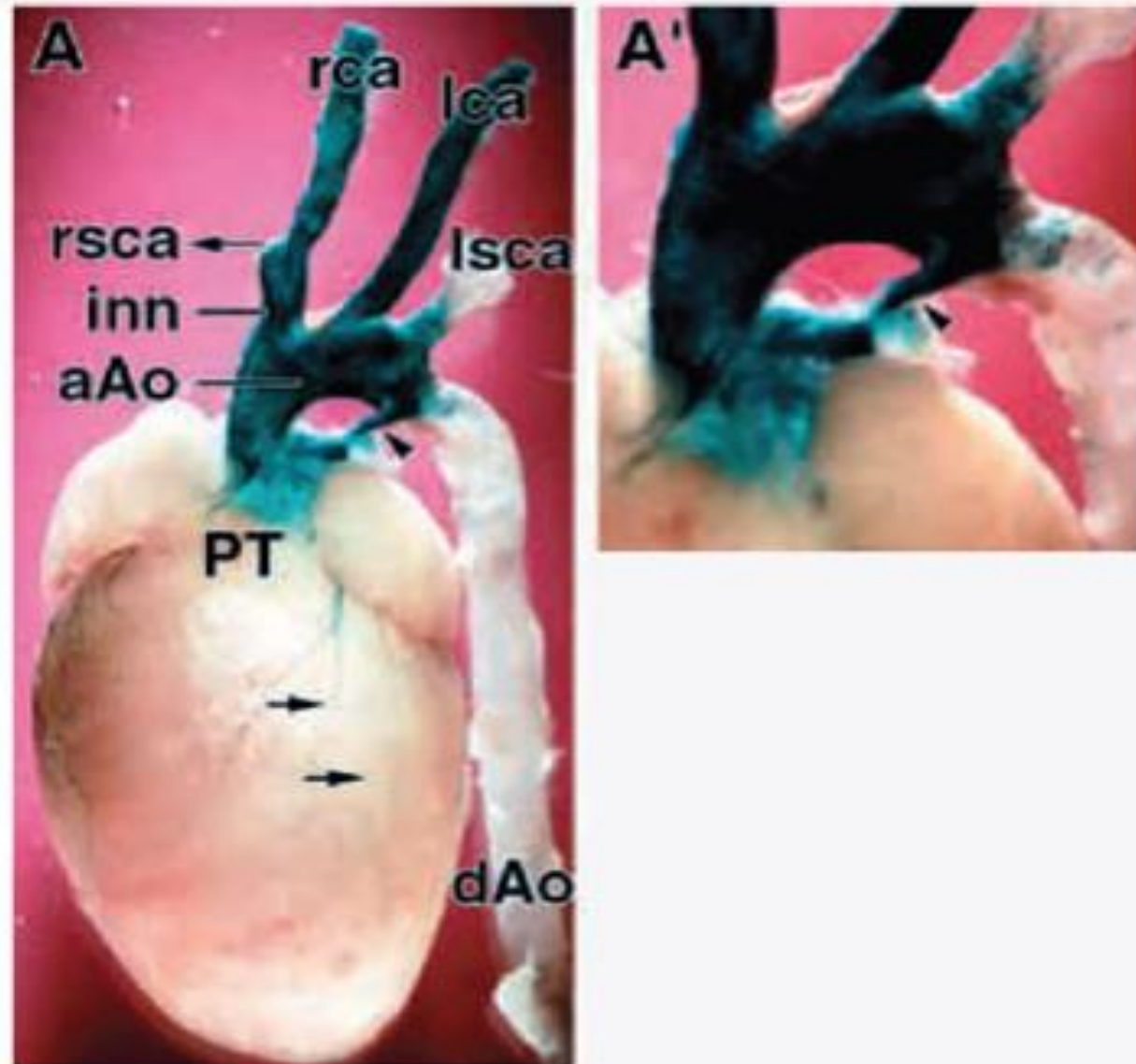
Partitioning of bulbus cordis and truncus arteriosus

CONOTRUNCAL region (outflow tract)

septation: - transformation of the myocardial outflow tract into the separated pulmonary trunk and aorta requires the cellular contribution of a distinct precursor pool called the **neural crest cells**.



Lineage Label Neural Crest Cells



Development of cardiac valves

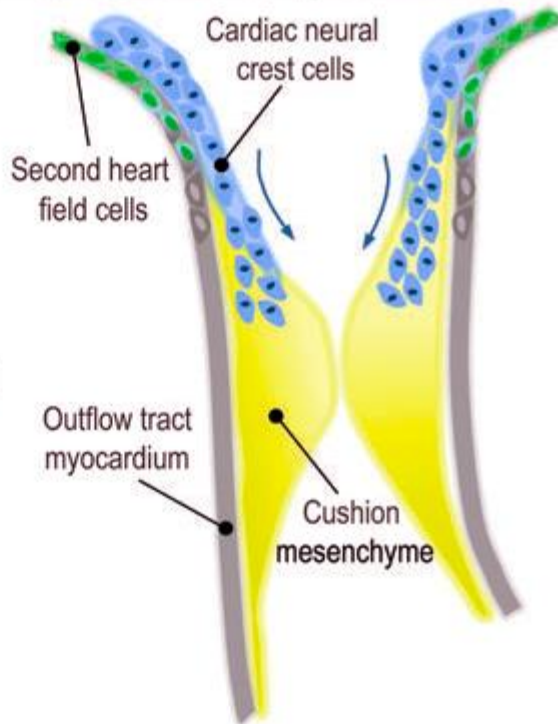
Cardiac neural crest cells



Anterior SHF

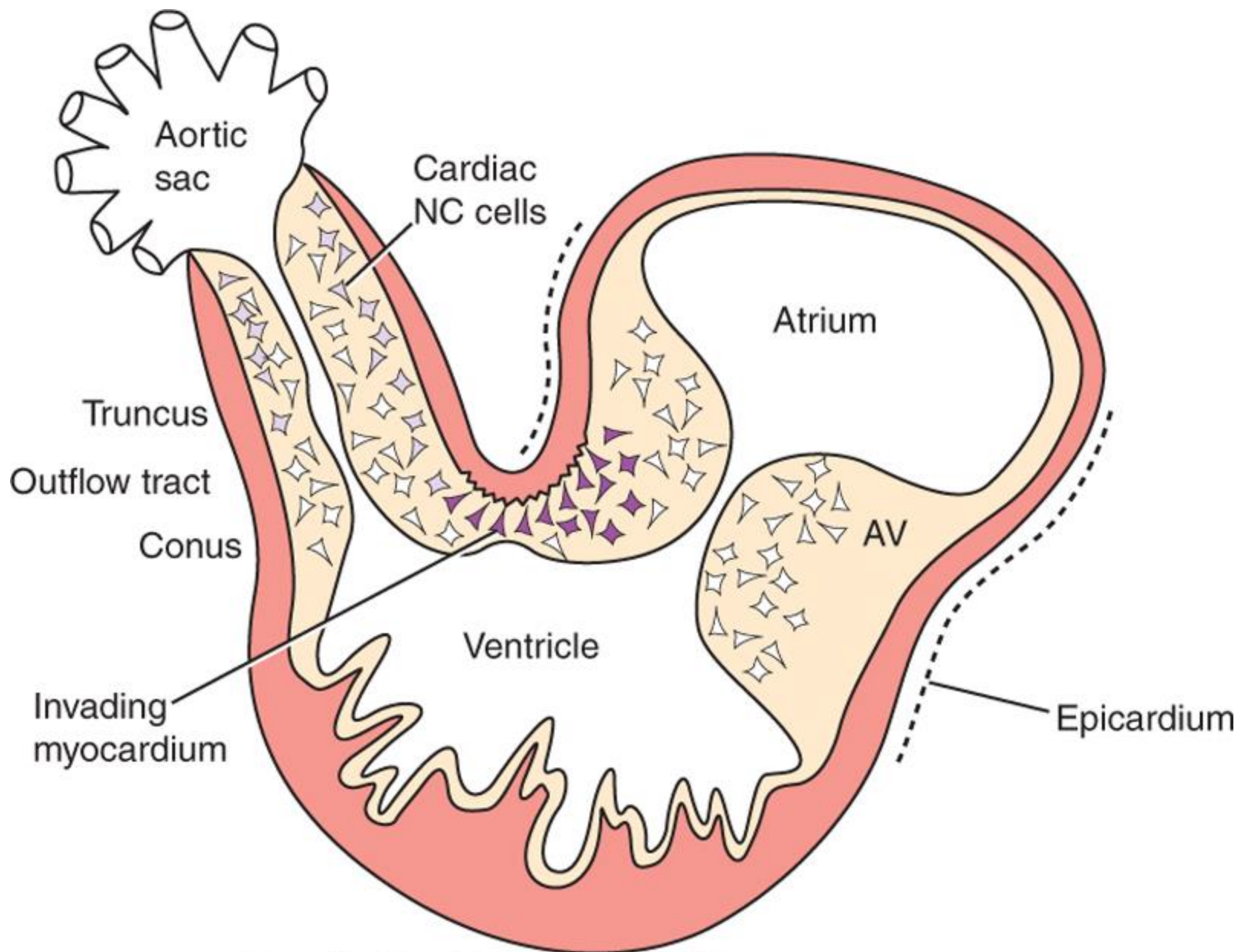


Cardiac neural crest migration into the heart



-**neural crest** cells invade the proximal and distal **outflow tract cushions** and contribute to the formation:

- semilunar valves**
- tunica media layer of the smooth muscular wall of the great vessels, aortic arch and the proximal carotids

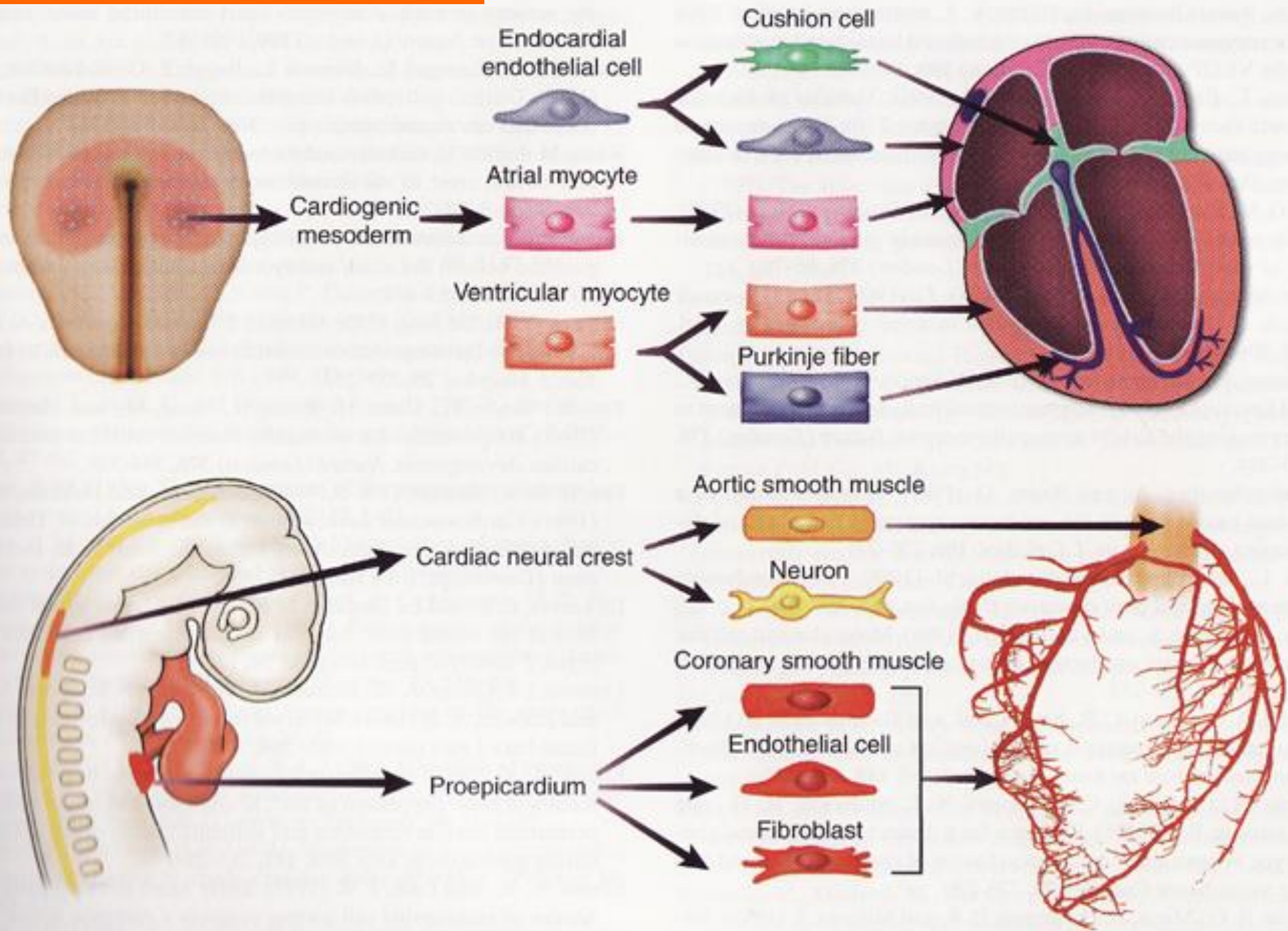


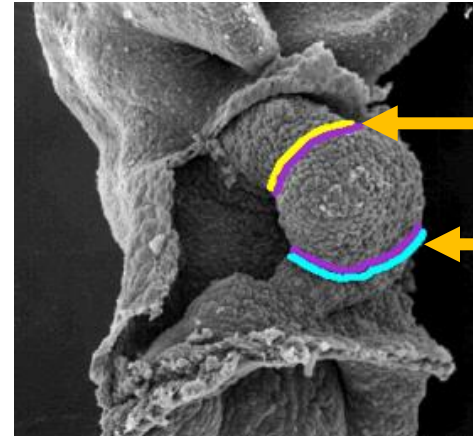
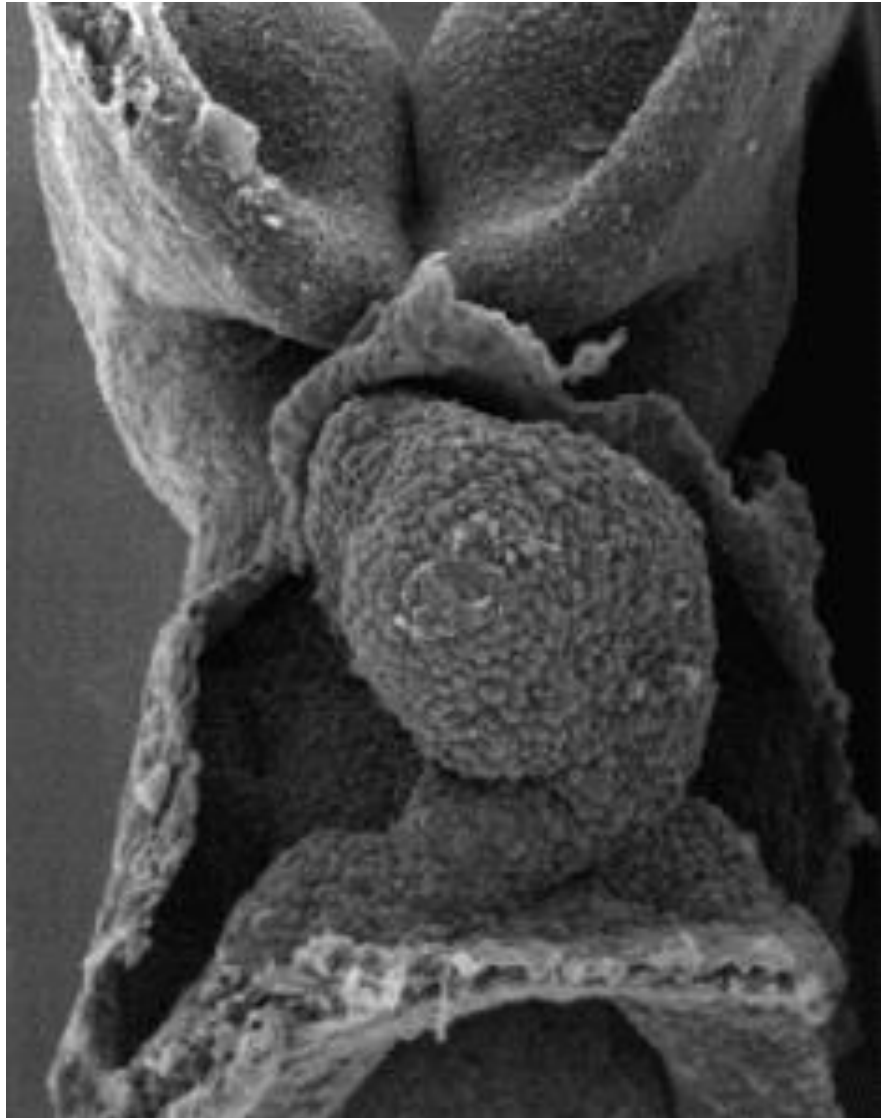
Development of cardiac valves - 4 distinct lineages

In addition to **secondary heart field lineage**, cardiac valves appear to be derived from at least **four distinct lineages**.

- 1. mesenchyme**, is derived from the **endocardium**, which has been confirmed by the transformation of endocardial endothelium to mesenchyme (ETM)
 - give rise to components of the major midline cushions and AV valve leaflets,
 - chordae tendineae
- 2. epicardium** and epicardial-derived cells (EPDCs)
 - epicardial-contribute not only to the annulus fibrosis and AV sulcus/groove, but also to the **mural mitral and tricuspid leaflets**
- 3. neural crest cells** contributes to the development
 - **AV septal complex** and the **AV valves**
 - neural crest cells do, in fact, migrate into the **septal tricuspid leaflet** and the **aortic leaflet of the mitral valve** while sparing the mural leaflets
 - penetrate through the sinus venosus mesocardium and proceed to the AV canal
- 4. cells of a bone marrow hematopoietic stem cell** origin are able to contribute to **valvular interstitial cells**

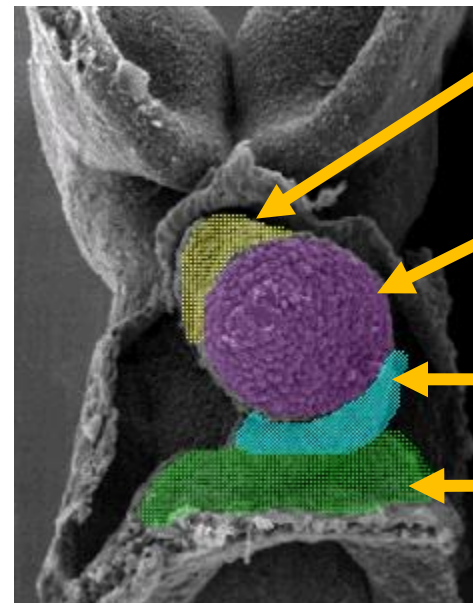
Heart-forming cells of the embryo





Sulcus
bulboventricularis

Sulcus
atrioventricularis

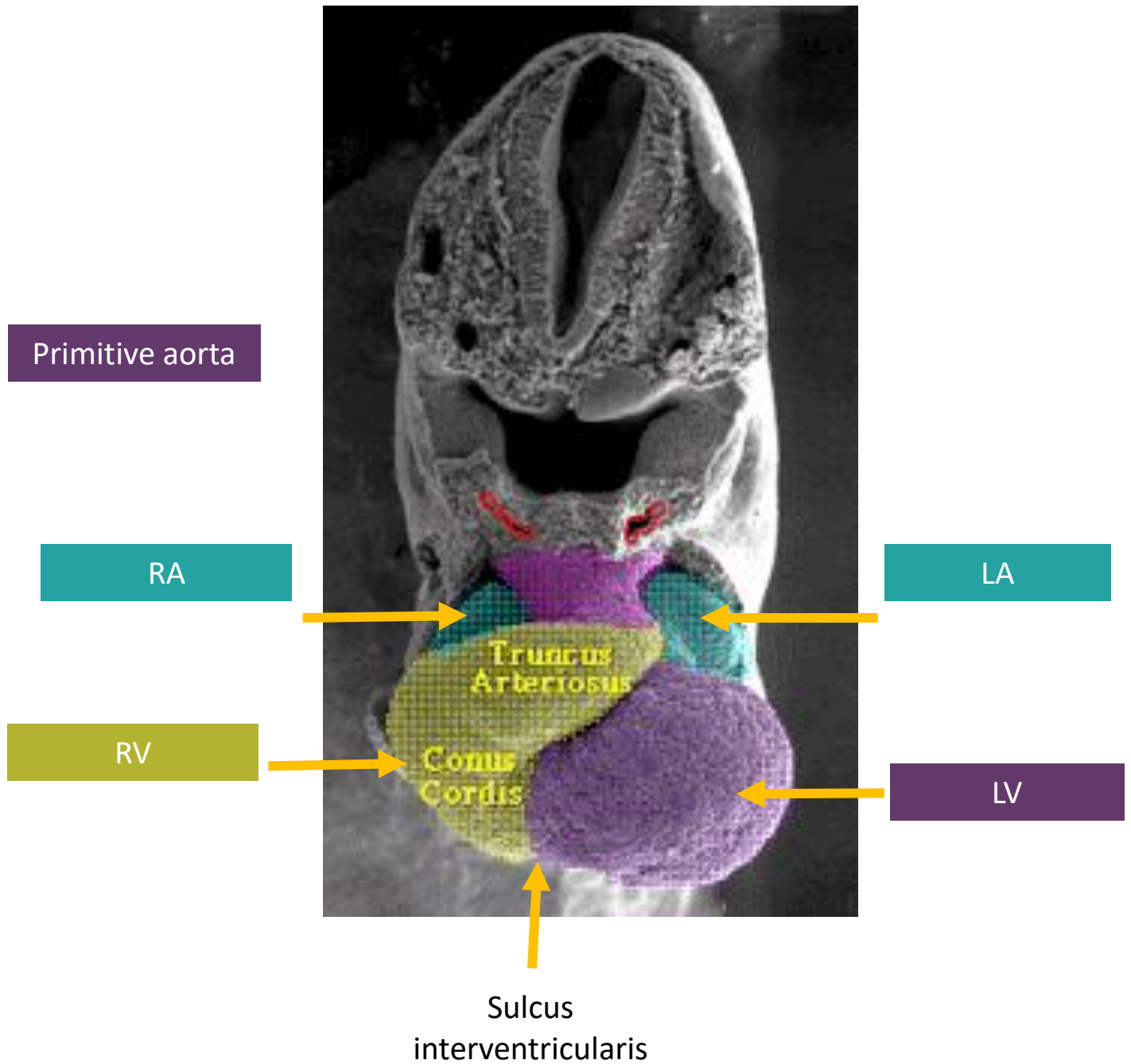


Truncus
arteriosus
+
Bulbus cordis

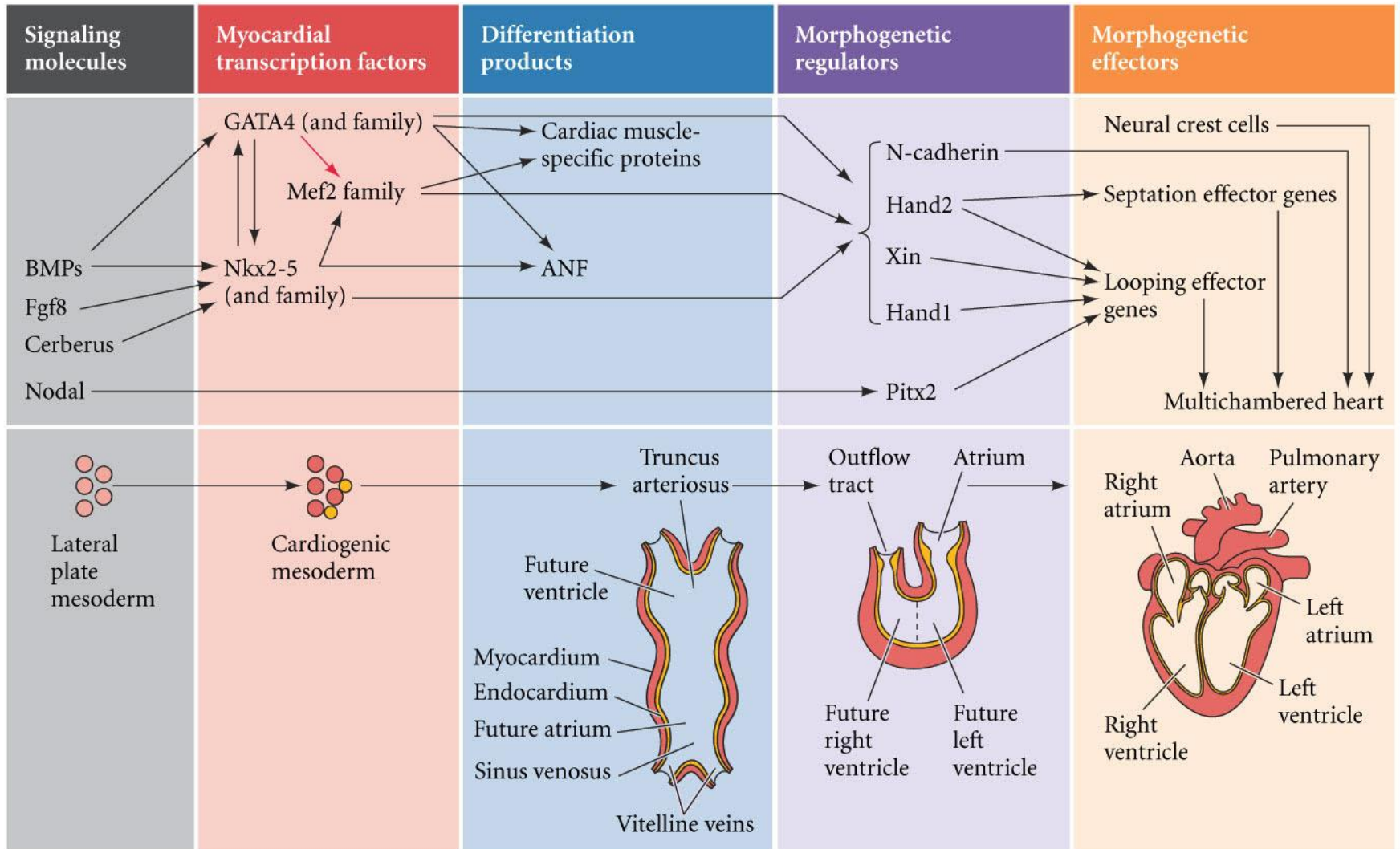
Ventriculus
communis

Atrium
commune

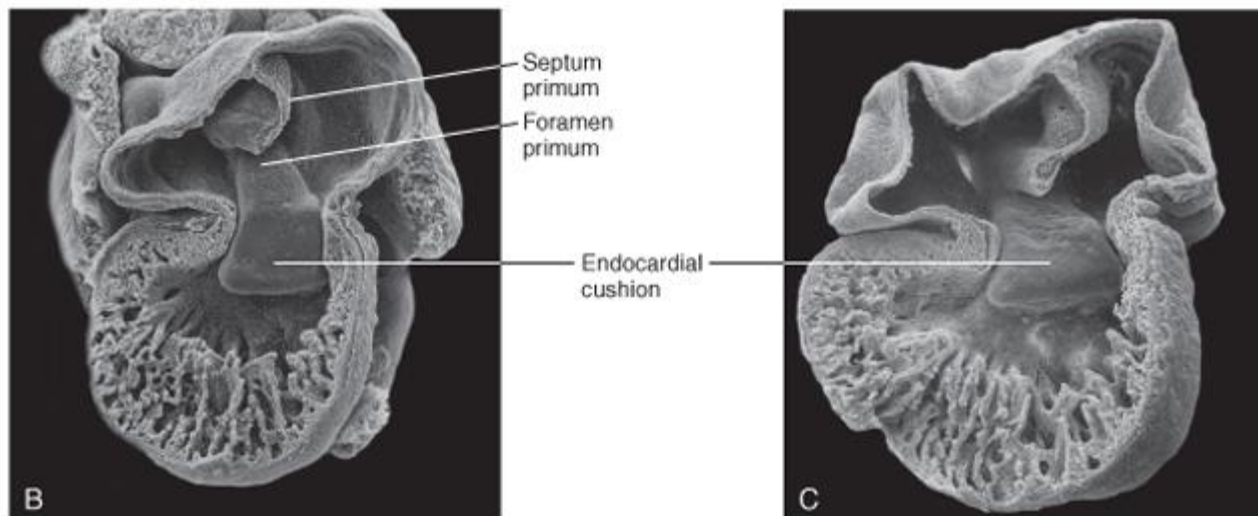
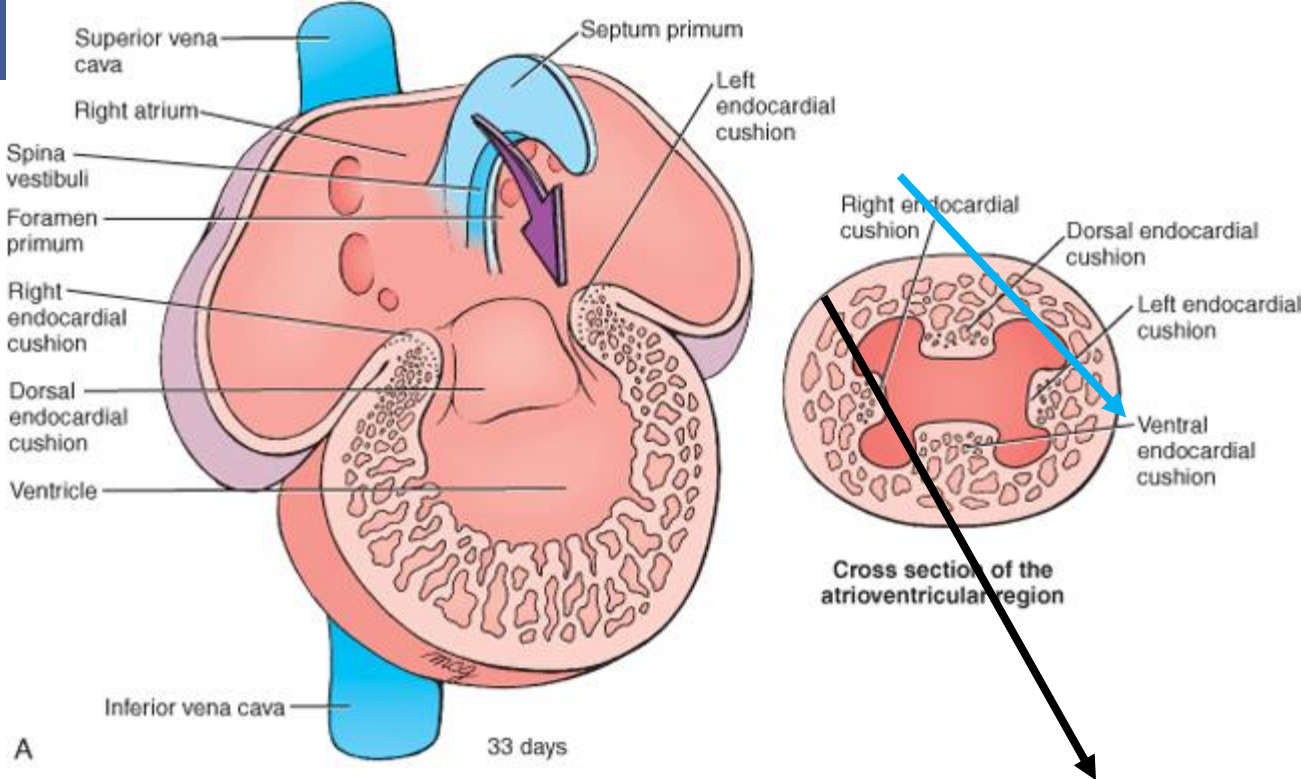
Sinus venosus



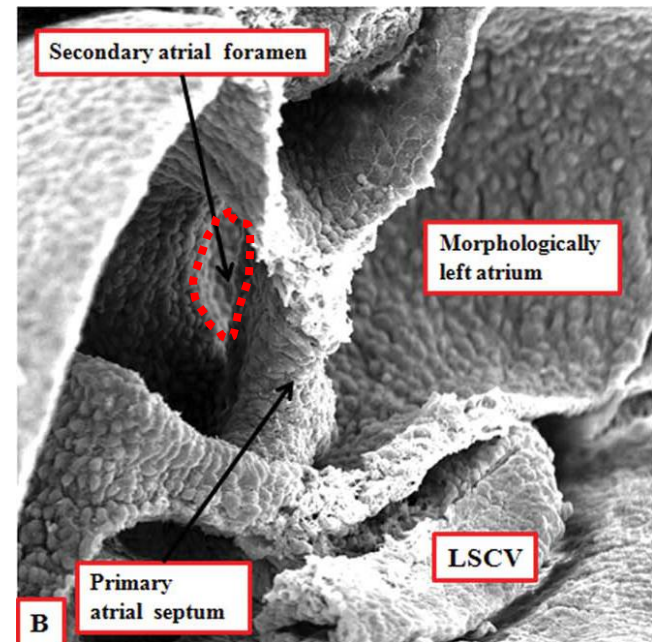
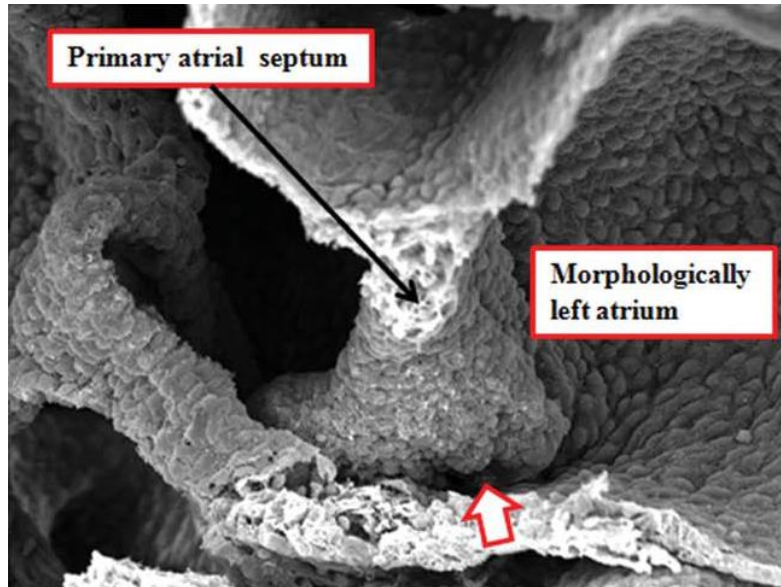
Summary of heart development



Formation of interatrial septum



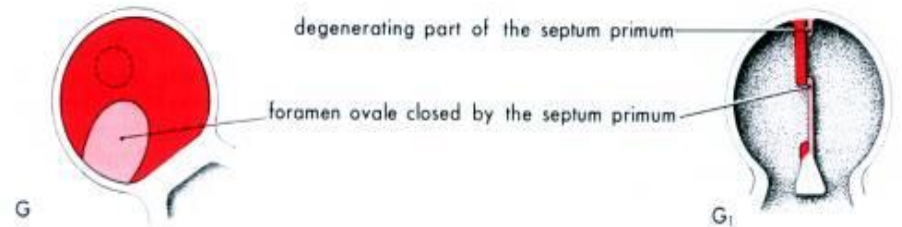
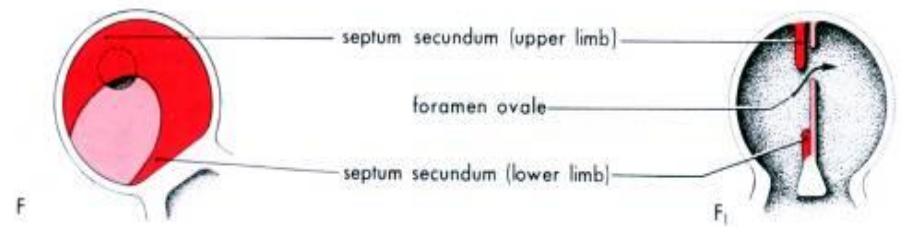
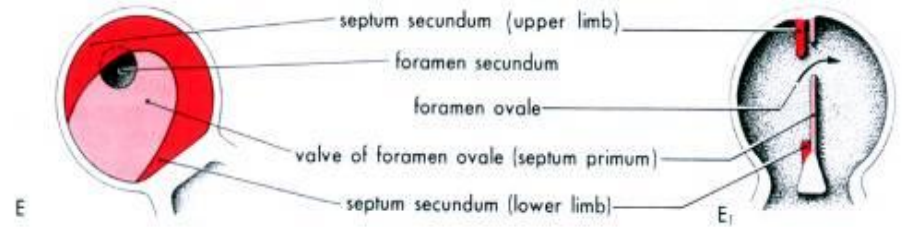
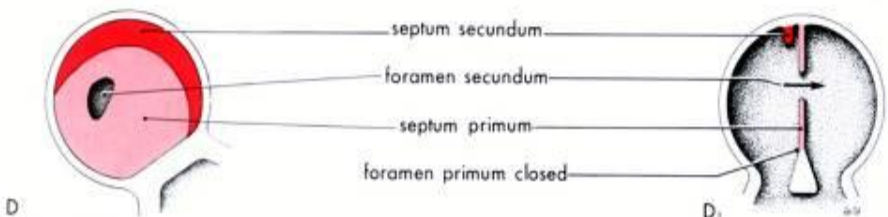
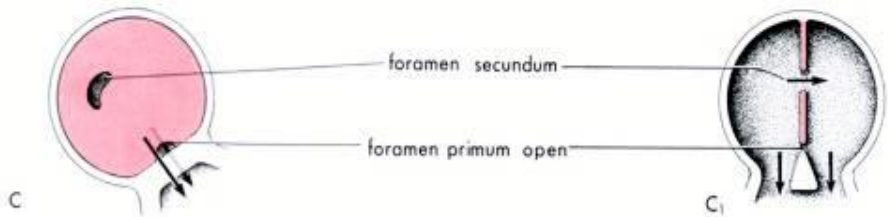
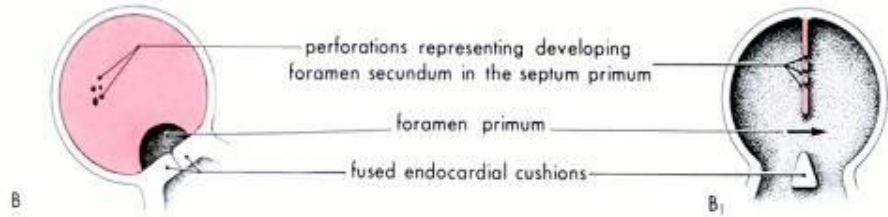
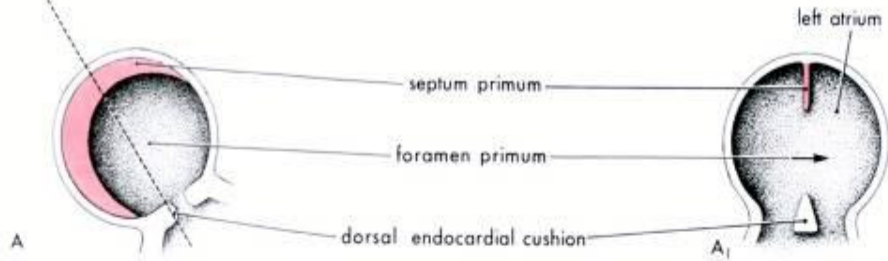
Septum primum – ostium primum – foramen secundum

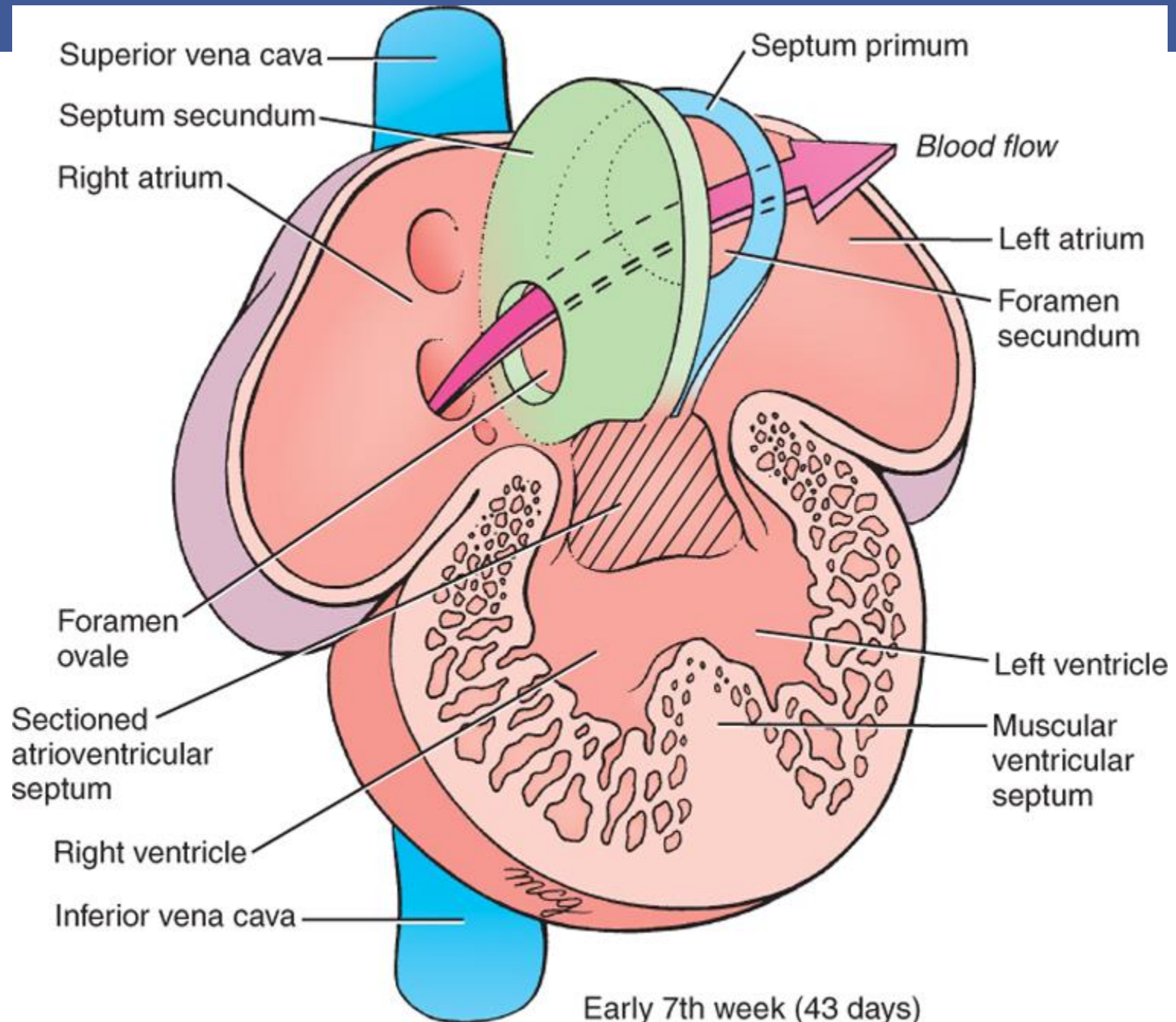


- primary atrial septum fuses with the atrial surface of atrioventricular endocardial cushions
- it is fusion of the mesenchymal tissues with the atrioventricular endocardial cushions that obliterates the primary atrial foramen.
- anchors the primary atrial septum to the atrioventricular cushions

-primary atrial septum grows as a muscular partition from the roof of the primary atrium
-this muscular shelf grows into the atrial cavity **between the systemic and pulmonary venous openings**
-originates from the part of the primary atrium that has molecularly **left characteristics**: expression of the *Pitx2* gene

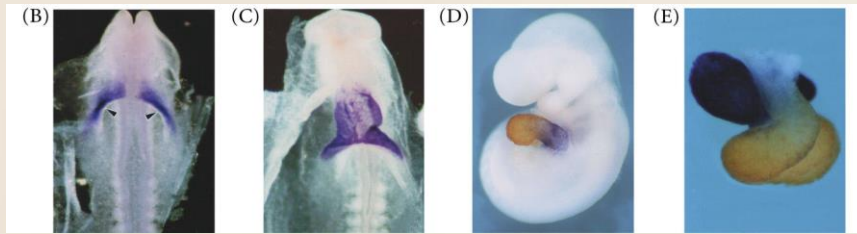
plane of sections A₁ to H₁





Congenital heart defects

- **Patent Ductus Arteriosus (10%)**
- **Atrial Septal Defect (10%)**
- **Ventricular Septal Defect (30%)**
- **Tetralogy of Fallot (6%)**
- **Transposition of the Great Arteries (4%)**
- **Coarctation of the Aorta**
- **Anomalous Venous Return**
- **Truncus Arteriosus**
- **Hypoplastic Left-Heart Syndrome**

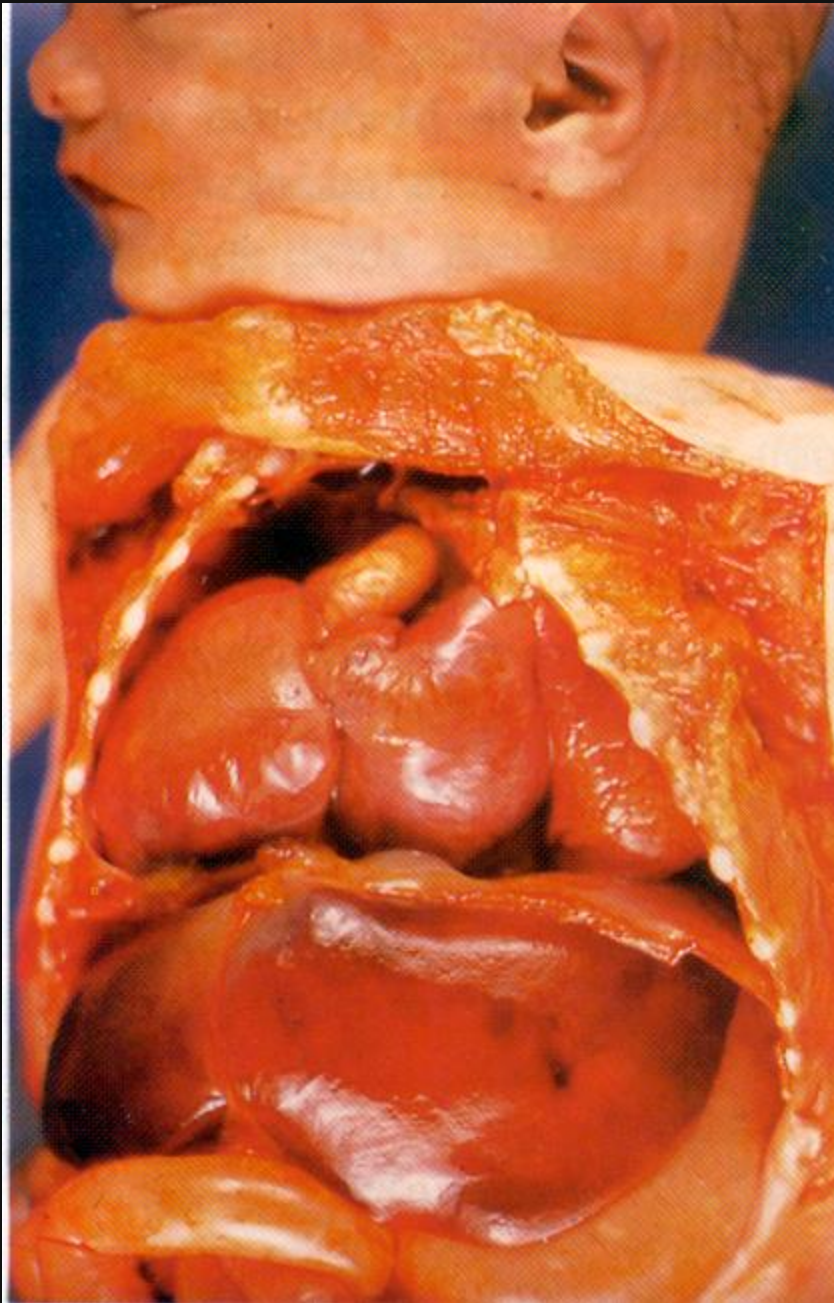




Holt-Oram syndrome:

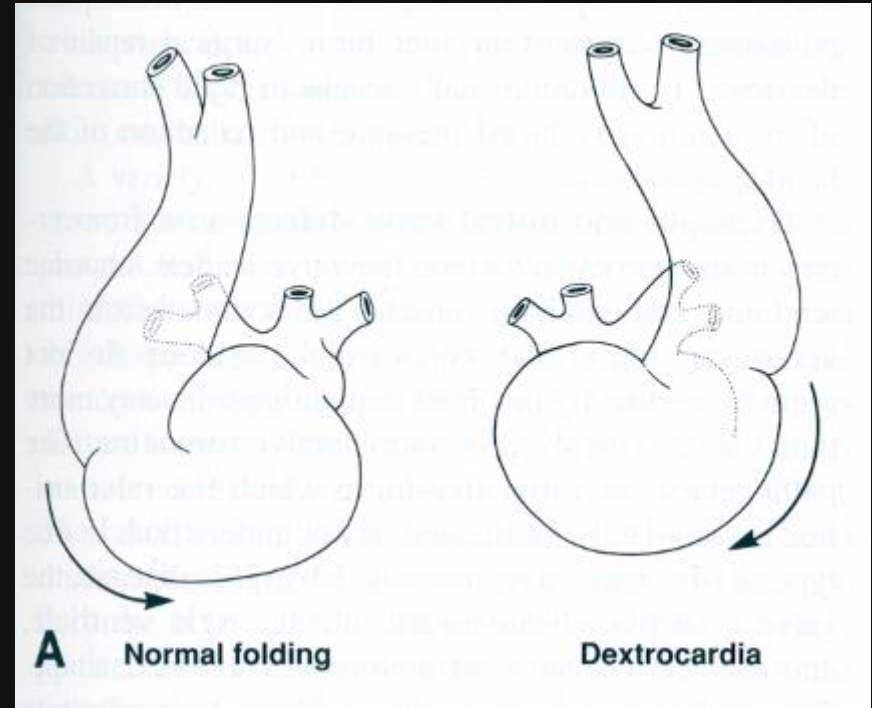
- TBX5 mutation
- 1:100.000
- limb malformation (100%)
and heart developmental defects (67%)
- phocomelia
- absence of the radius



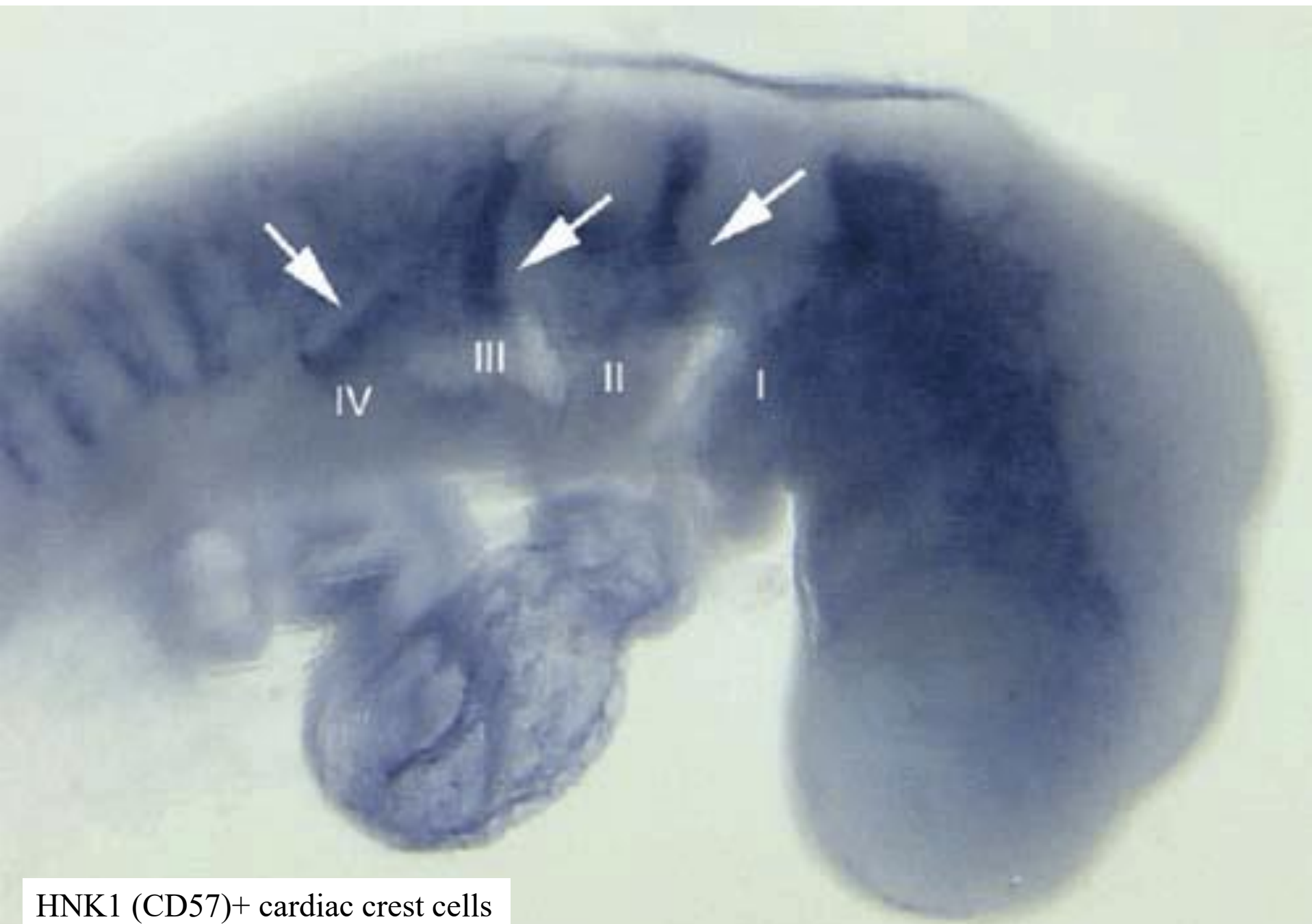


Dextrocardia; 1:10.000

(Shh, activin IIa receptor mutation)



Ectopia cordis:



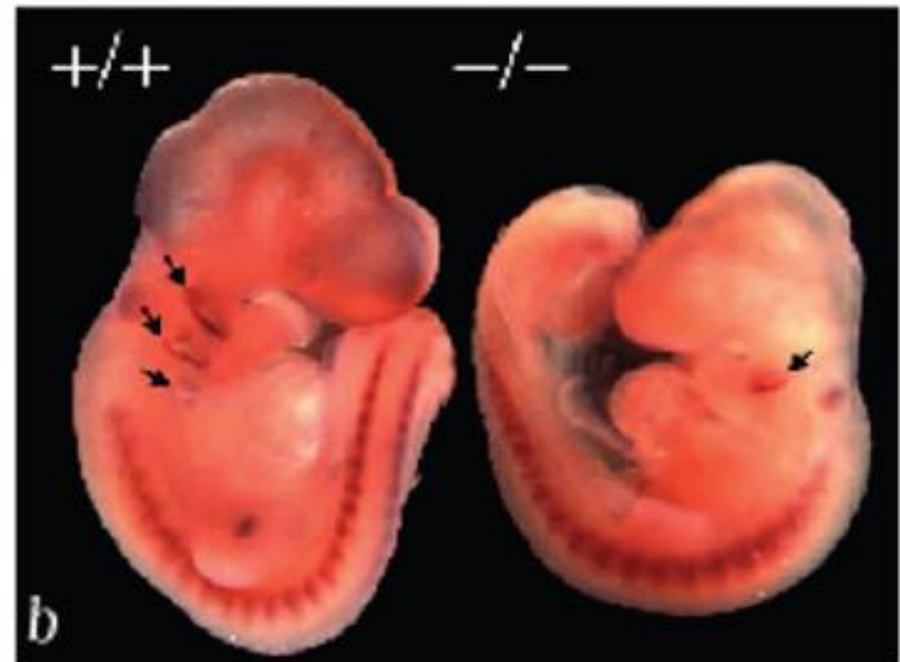
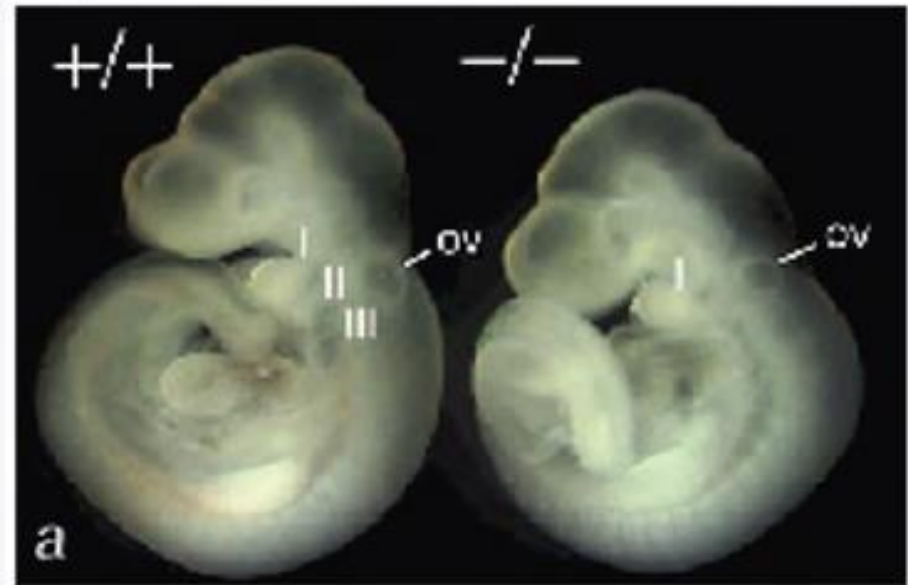
DiGeorge Syndrome

Defects in cranial and vagal neural crest; glandular deficiencies, craniofacial abnormalities, heart defects.

-aplasia of thymus and parathyroid gland

Mutations in *Tbx1* (transcription factor) is responsible for this syndrome.

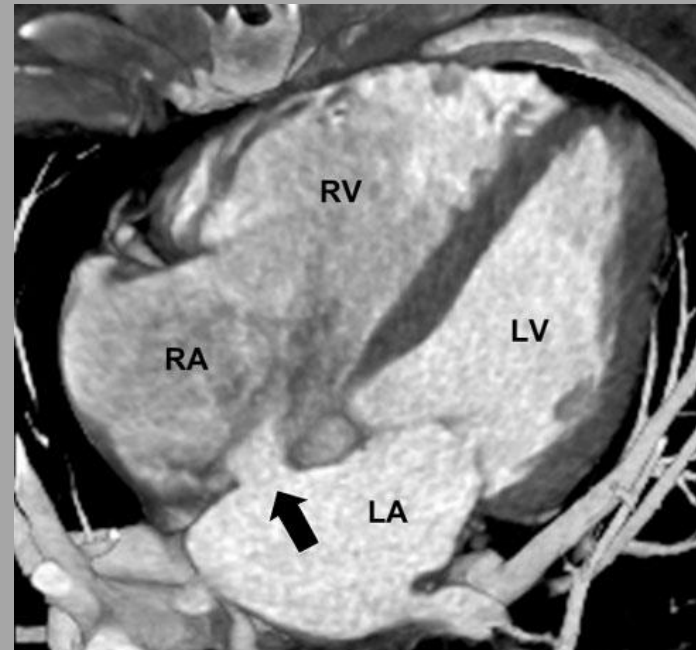
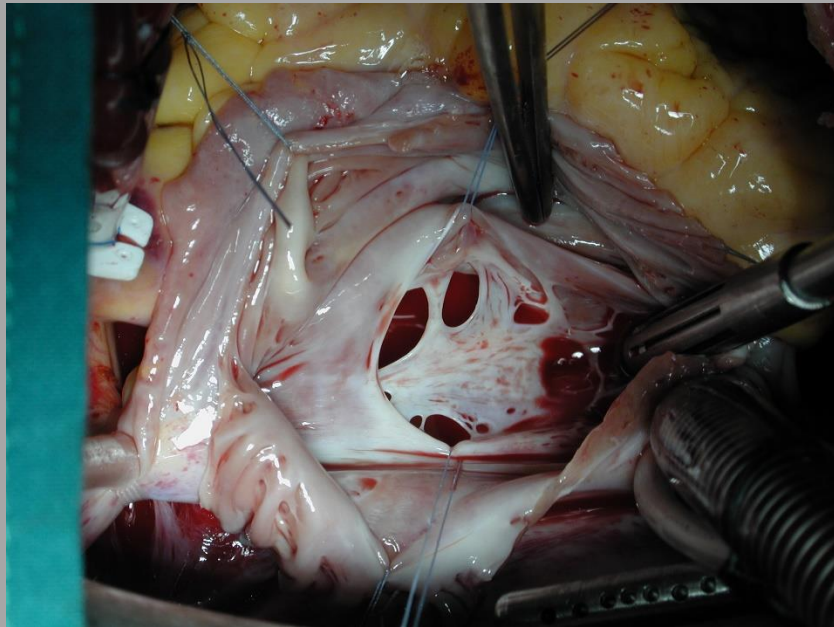
is associated with a deletion on chromosome 22, is characterized by hypoplasia and reduced function of the thymus, thyroid, and parathyroid glands and cardiovascular defects, such as persistent truncus arteriosus and abnormalities of the aortic arches.



Atrial Septal Defect

(6-10% of all congenital heart diseases)

-foramen ovale allows blood to bypass the nonfunctional fetal lungs while the fetus obtains its oxygen from the placenta

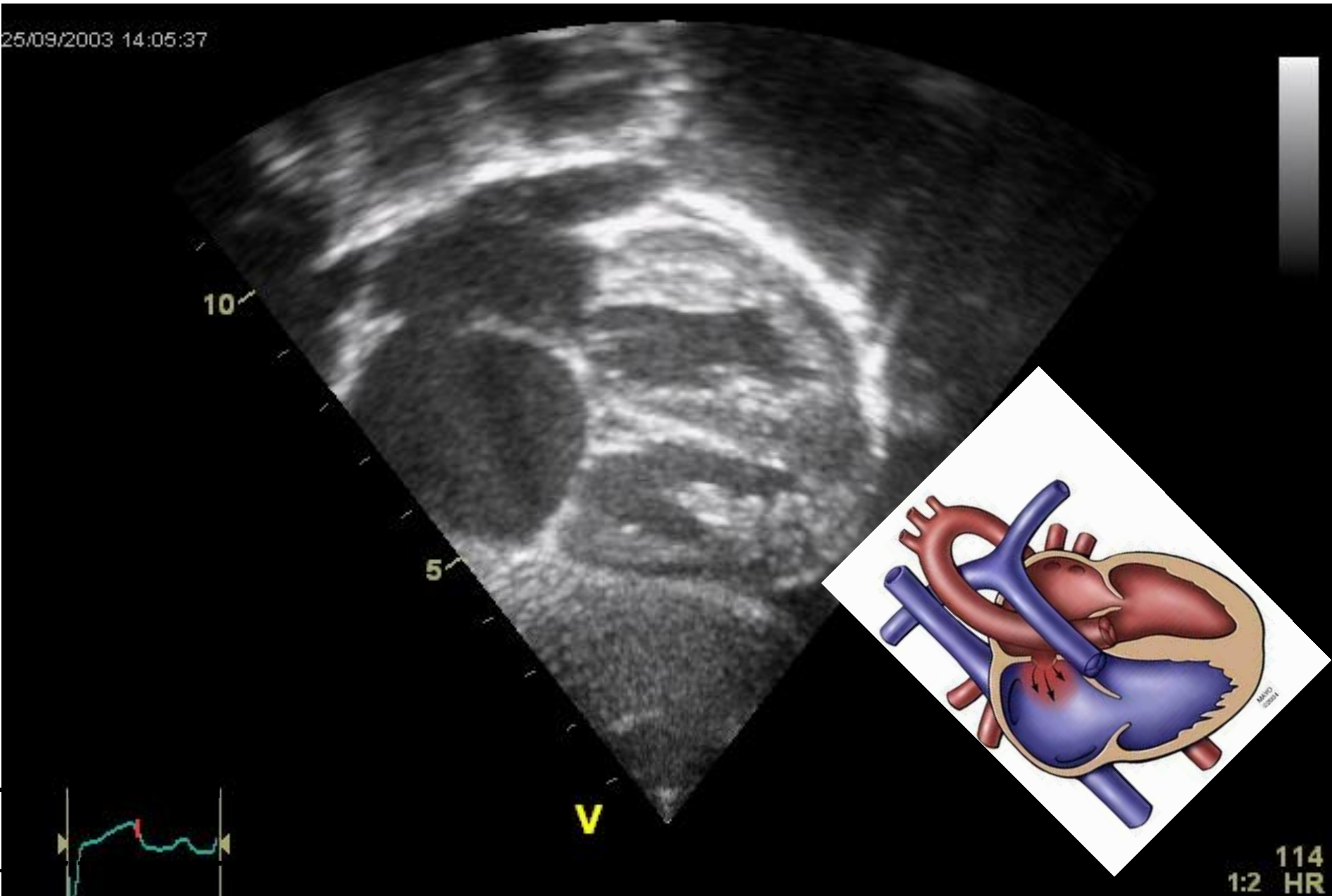


Catheter procedure - percutaneous closure safer

ATRIAL SEPTAL DEFECTS (ASD)

- Three types exist : primum, secundum and sinus venosus
- The most common is the *secundum* type
- Symptoms: *None* in childhood, arrhythmias in the 3 rd decade

atrial septal defect, ASD 1:1.500

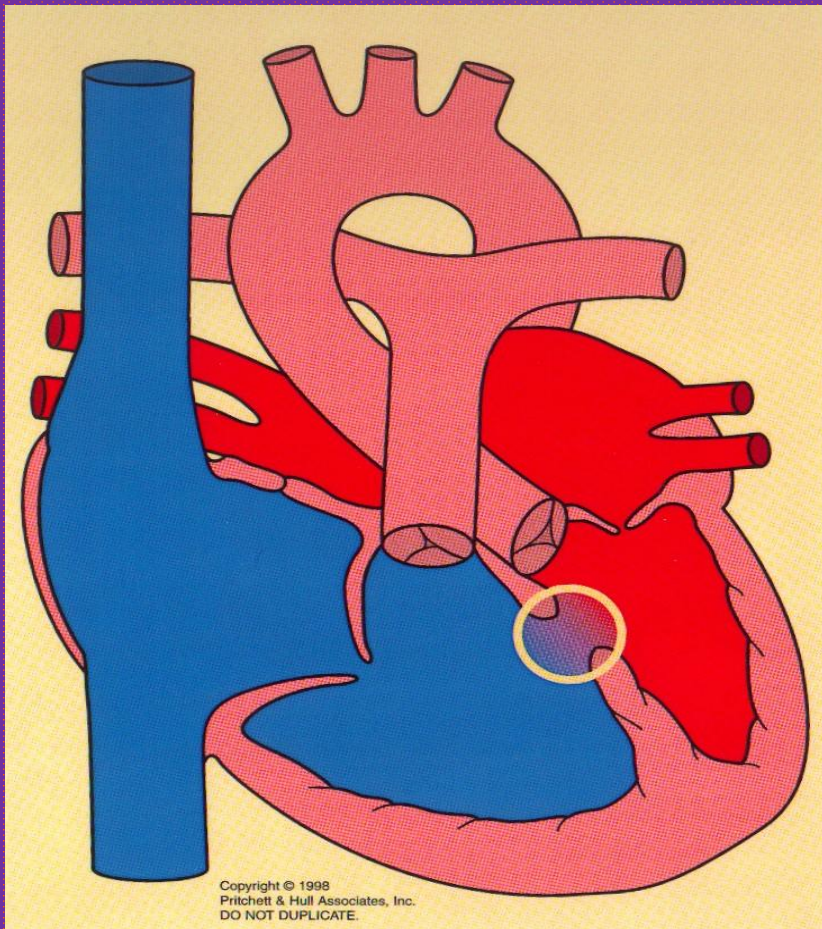


VENTRICULAR SEPTAL DEFECTS (VSD; 30%)

- This is the most common form of CHD
- The VSDs are subdivided according to the part of the septum they occur in :
Muscular, perimembranous, inlet, outlet
- A large VSD causes left ventricular enlargement
- With a small VSD there is normal growth and development

VENTRICULAR SEPTAL DEFECTS

2-6:1.000



There has to be a
RIGHT to LEFT shunt
to cause cyanosis

Tetralogy of Fallot (5-9:10.000)

- Most common cyanotic heart disease.
- The 4 abnormalities include:
 - Pulmonary stenosis
 - RVH
 - VSD
 - Overriding Aorta
- Signs include cyanosis, murmur, squatting and the knee chest position which increases aortic wave reflection, increasing pressure on the left side of the heart, decreasing the right to left shunt thus decreasing the amount of deoxygenated blood entering the systemic circulation) and spells (acute hypoxia (tet spells), characterized by sudden cyanosis and syncope).

Tetralogy of Fallot

Four abnormalities that results in insufficiently oxygenated blood pumped to the body

1. Narrowing of the pulmonary valve

3. Displacement of aorta over ventricular septal defect

4. Ventricular septal defect- opening between the left and right ventricles

2. Thickening of wall of right ventricle

Overriding
aorta

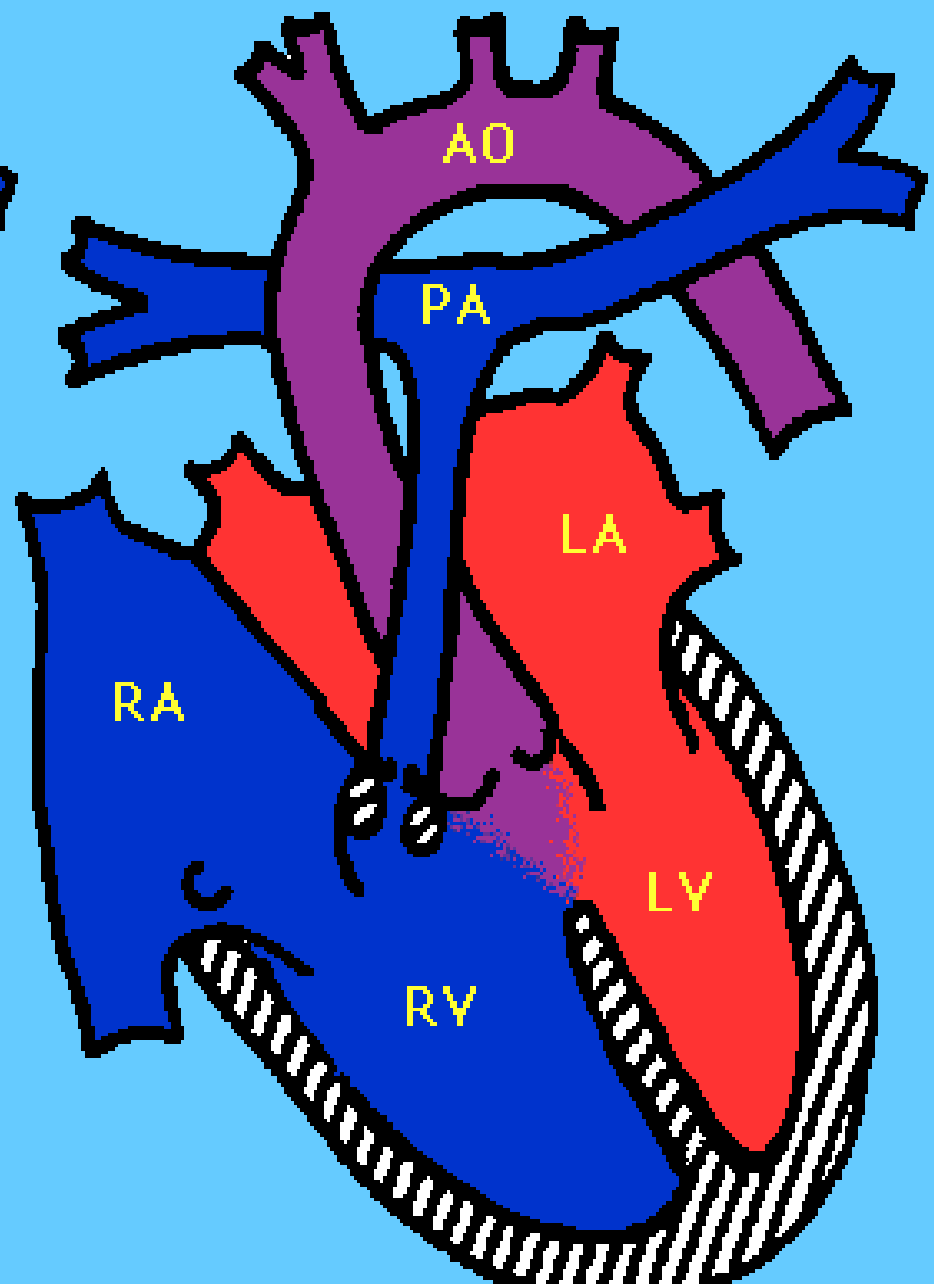
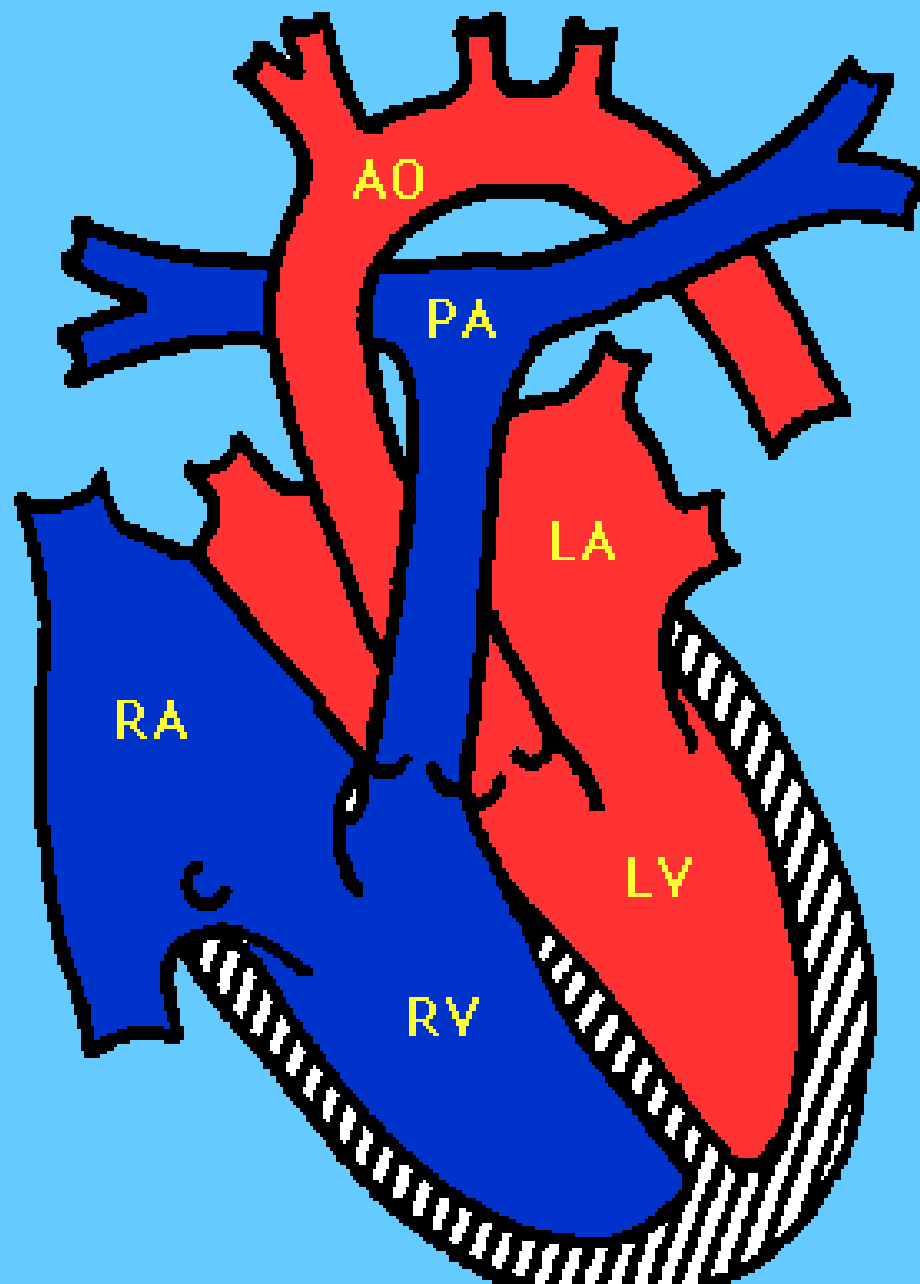
Enlarged
right
ventricle

Stenotic
pulmonary
trunk



B

Tetralogy of Fallot



TOF cont..

- holding the baby in a knee chest position
- Morphine
- Oxygen, beta blocker, general anesthesia,

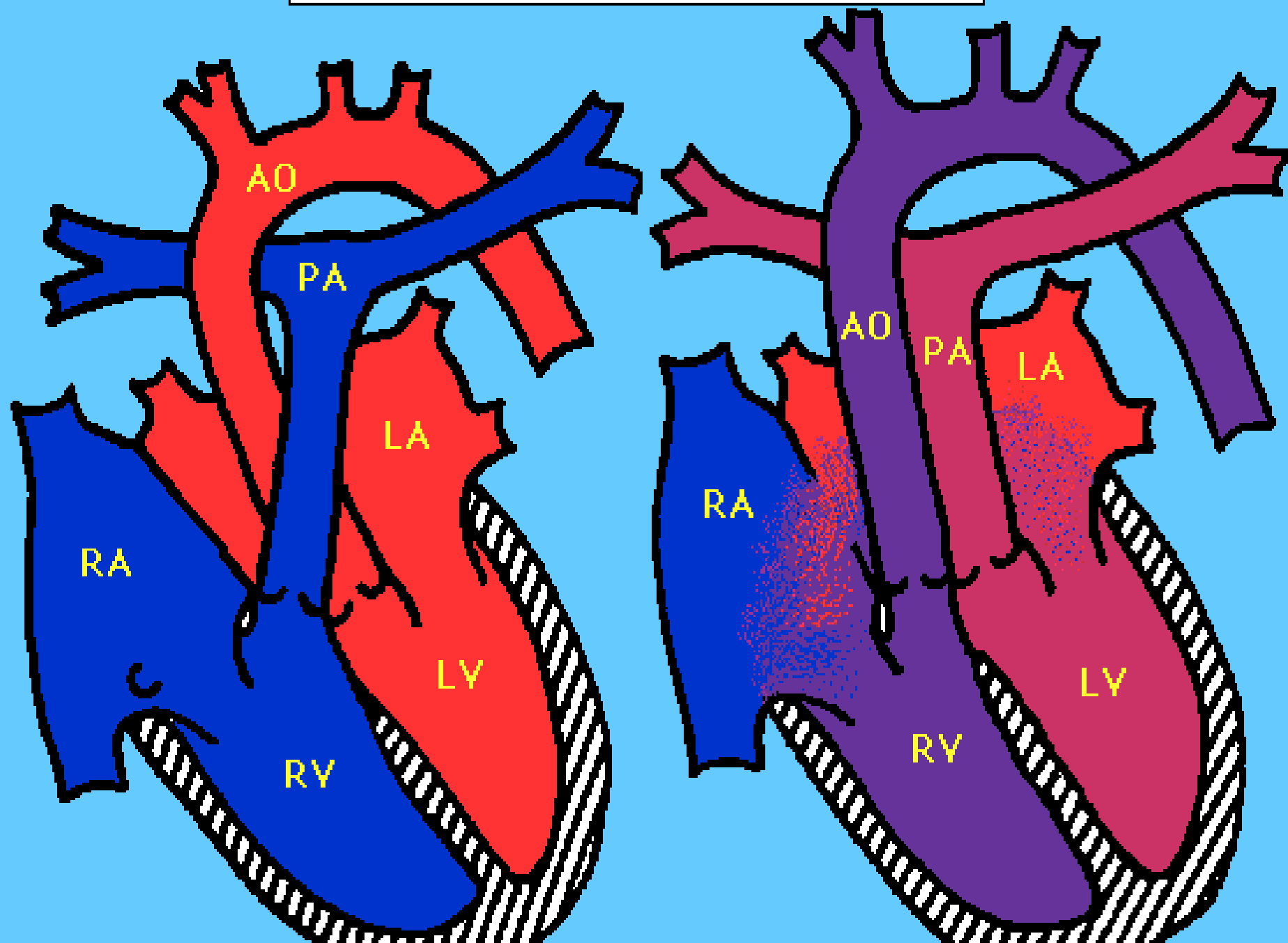
Children with Tetralogy of Fallot exhibit bluish skin during episodes of crying or feeding.



Tet spells can be spontaneous or triggered by certain events such as prolonged crying and episodes of exertion such as during feeding or even pooing.

Toddlers and older children may prefer to squat. This helps increase blood flow to the lungs.

Transposition of the Great Vessels



Transposition of the great Arteries

- The aorta arises from the right ventricle and the pulmonary artery from the left.
- The mixing of the blood occurs at the PFO and the PDA.
- The signs include cyanosis and cardiomegaly.
Reverse differential cyanosis!
- There may be no murmur.
- An echocardiogram is diagnostic.